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Inventor(s)	McCullough; Adam B.

Drug delivery device with proximity sensor

Abstract

A drug delivery device, in the form of an injector, may include one of a number of systems for limiting the delivery of a medical fluid or drug product in case of movement of (e.g., removal of) the injector relative to the patient as determined by a proximity sensor. The drug delivery system may in the alternative or in addition include systems for indicating the amount of medical fluid or drug product delivered (or not delivered) in case of movement of (e.g., removal of) the injector relative to the patient as determined by the proximity sensor. The injector may be, for example, an on-body injector or an hand-held autoinjector.

Inventors:	McCullough; Adam B. (Westlake Village, CA)
Applicant:	AMGEN INC. (Thousand Oaks, CA)
Family ID:	1000008821577
Assignee:	AMGEN INC. (Thousand Oaks, CA)
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Primary Examiner: Ponton; James D

Attorney, Agent or Firm: MARSHALL, GERSTEIN & BORUN LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. Ser. No. 16/942,733, filed Jul. 29, 2020, which is a continuation of U.S. patent application Ser. No. 15/047,853, filed Feb. 19, 2016, which is a continuation of PCT/US2015/066597, filed Dec. 18, 2015, which claims priority to U.S. Provisional Patent Application No. 62/094,395, filed Dec. 19, 2014, the entire contents of each of which are hereby incorporated by reference.

FIELD OF THE DISCLOSURE

(1) This disclosure is directed to a drug delivery device and, in particular, to a drug delivery device that limits drug delivery in accordance with the state of a proximity sensor.

BACKGROUND

(2) Medical fluids and drug products (e.g., drugs) can be administered to a patient through the use of drug delivery devices, such as autoinjectors or on-body injectors or infusers. These devices may replace delivery systems using the combination of a syringe and a vial of the medical fluid or drug product or a pre-filled syringe. Autoinjectors and on-body injectors may be used to automate the needle or cannula insertion and the drug delivery or administration process, thereby simplifying the process for certain patient groups or sub-groups for which use of the syringe/vial combination or pre-filled syringe systems would be disadvantageous, such as groups or sub-groups demonstrating physiological or psychological impediment.

(3) In some instances, after automated insertion of the needle or cannula, the autoinjectors and on-body injectors may continue to allow drug to pass through the needle or cannula even though a component of the device (e.g., injector) has become detached from the patient and the needle or cannula no longer is inserted into the patient. This has several consequences. First, the patient does not receive the full dose of the drug, which may have a negative effect on the patient. Second, the patient may not be aware of the fact that the full dose of the drug has not been delivered, leaving the patient with the false impression that the entire dose has been delivered when it has not. Third, even if the patient is aware of the fact that the full dose has not been delivered, the patient may be unable to determine just how much of the dose was delivered. Conversely, the patient may not be able to determine how much of the dose was not delivered.

(4) As set forth in more detail below, the present disclosure provides an improved drug delivery device embodying advantageous alternatives to the conventional devices and methods.

SUMMARY

(5) According to an aspect of the disclosure, a drug delivery device can include a reservoir, a cannula, a drive, a lock, a housing and a proximity sensor. The reservoir can include a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends. The cannula can have an operational state wherein the cannula is connected in fluid communication with the reservoir. The drive can be coupled to the plunger assembly to move the plunger between the first and second ends. The lock can be configured to selectively limit movement of the plunger between the first and second ends of the reservoir. The reservoir, the drive, and the lock can be disposed at least partially within the housing. The proximity sensor can be operably coupled to move relative to the housing, and can have a first sensor state wherein the proximity sensor extends from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing relative to the first sensor state. The lock can limit movement of the plunger assembly when the proximity sensor is in the first sensor state.

(6) According to a further aspect of the disclosure, a method of limiting the delivery of a drug from a drug delivery device after the drug delivery device is removed from a patient can be provided. The method can include (a) providing a drug delivery device, wherein the drug delivery device can

include a reservoir, a cannula, a drive, a lock, a housing and a proximity sensor. The reservoir can include a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends. The cannula can have an operational state wherein the cannula is connected in fluid communication with the reservoir. The drive can be coupled to the plunger assembly for moving the plunger between the first and second ends. The lock can be configured to selectively limit movement of the plunger between the first and second ends of the reservoir. The reservoir, the drive, and the lock can be disposed at least partially within the housing. And the proximity sensor can be operably coupled to move relative to the housing, and can have a first sensor state wherein the proximity sensor extends from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing relative to the first sensor state. The lock can limit movement of the plunger assembly when the proximity sensor is in the first sensor state. The method can further include, (b) when the drug delivery device is removed from the patient, causing the proximity sensor to occupy the first sensor state after having occupied the second sensor state. And, the method can include, (c) transmitting information from the proximity sensor to the lock to indicate that the proximity sensor has moved to the first sensor state. Finally, the sensor can include (d) causing the lock to limit movement of the plunger assembly upon receiving information that the proximity sensor is in the first sensor state.

(7) According to another aspect of the present disclosure, a drug delivery device such as an injector can include a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends, and a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir. The injector also includes a spring coupled to the plunger assembly to move the plunger between the first and second ends, a lock selectively coupled to one of the plunger assembly and the spring to limit movement of the plunger between the first and second ends with the lock coupled to the one of the plunger assembly and the spring, a housing, the reservoir, spring, and lock disposed within the housing, and a proximity sensor coupled to the lock and moveable relative to the housing, the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the lock coupled to the one of the plunger assembly and the spring with the proximity sensor in the first sensor state.

(8) According to another aspect of the present disclosure, a drug delivery device such as an injector can include a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends, and a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir. The injector also includes a gas source having an operational state wherein the gas source is in fluid communication with the plunger to move the plunger between the first and second ends, a lock comprising a vent selectively coupled to the gas source to limit movement of the plunger between the first and second ends, a housing, the reservoir, gas source, and lock disposed within the housing, and a proximity sensor coupled to the lock and moveable relative to the housing, the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the vent coupled to the gas source with the proximity sensor in the first sensor state.

(9) According to a further aspect of the present disclosure, a drug delivery device such as an injector can include a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends, and a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir. The injector also includes a spring coupled to the plunger assembly to move the plunger between the first and second ends, an indicator selectively mechanically coupled to the plunger assembly, a housing, the reservoir, spring, and indicator at least partially disposed within the housing, and a proximity sensor coupled to the indicator and moveable relative to the housing,

the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the indicator coupled to the plunger assembly with the proximity sensor in the second sensor state and the indicator decoupled from the plunger assembly with the proximity sensor in the first sensor state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure will be more fully understood from the following description taken in conjunction with the accompanying drawings. Some of the figures may have been simplified by the omission of selected elements for the purpose of more clearly showing other elements. Such omissions of elements in some figures are not necessarily indicative of the presence or absence of particular elements in any of the exemplary embodiments, except as may be explicitly delineated in the corresponding written description. None of the drawings are necessarily to scale.
- (2) FIG. 1 is a cross-sectional view of an embodiment of a drug delivery device with a sliding lock that cooperates with a plunger arm and a proximity sensor in a first, off-body state;
- (3) FIG. 2 is a cross-sectional view of the embodiment of FIG. 1 taken along line 2-2, with the proximity sensor in the first, off-body state;
- (4) FIG. 3 is a cross-sectional view of the embodiment of FIG. 1 taken along line 2-2, with the proximity sensor in a second, on-body state;
- (5) FIG. 4 is a cross-sectional view of another embodiment of a drug delivery device with a pivoting lock that cooperates with a plunger arm and a proximity sensor in a first, off-body state;
- (6) FIG. 5 is a cross-sectional view of the embodiment of FIG. 4 taken along line 5-5, with the proximity sensor in the first, off-body state;
- (7) FIG. 6 is a diagram relating the motion of the ends of the lever of the embodiment illustrated in FIGS. 4 and 5;
- (8) FIG. 7 is a cross-sectional view of an embodiment of a drug delivery device with a sliding lock that cooperates with a spring and a proximity sensor in a first, off-body state;
- (9) FIG. 8 is a cross-sectional view of the embodiment of FIG. 7 taken along line 8-8, with the proximity sensor in a second, on-body state;
- (10) FIG. 9 is a cross-sectional view of an embodiment of a drug delivery device with a pivoting lock that cooperates with a spring and a proximity sensor in a first, off-body state;
- (11) FIG. 10 is a cross-sectional view of the embodiment of FIG. 9 taken along line 10-10, with the proximity sensor in a second, on-body state;
- (12) FIG. 11 is a cross-sectional view of an embodiment of a drug delivery device with a lock including a vent and a proximity sensor in a second, on-body state;
- (13) FIG. 12 is a cross-sectional view of the embodiment of FIG. 11 with the proximity sensor in a first, off-body state;
- (14) FIG. 13 is a perspective view of an injector with a fluid delivery indicator and a proximity sensor;
- (15) FIG. 14 is a partial cross-sectional view of the embodiment of FIG. 13 in an off-body state;
- (16) FIG. 15 is a partial cross-sectional view of the embodiment of FIG. 13 in the on-body state;
- (17) FIG. 16 is a perspective view of an on-body injector, which may incorporate any one of the foregoing embodiments illustrated in FIGS. 1-15;
- (18) FIG. 17 is a cross-sectional view of the on-body injector of FIG. 16;
- (19) FIG. 18 is a cross-sectional view of the on-body injector of FIG. 16;
- (20) FIG. 19 is a perspective view of an autoinjector, which may incorporate any one of the embodiments illustrated in FIGS. 1-15;

- (21) FIG. 20 is a cross-sectional view of the autoinjector of FIG. 19;
- (22) FIG. 21 is a cross-sectional view of an embodiment of a drug delivery device with a proximity sensor and lock including a movable plate in a first, off-body state;
- (23) FIG. 22 is a cross-sectional view of the embodiment of FIG. 21 with the proximity sensor and lock in a second, on-body state;
- (24) FIG. 23 is a cross-section view of an alternative version of the auto-injector of FIGS. 19 and 20 including a needle assembly that serves as a proximity sensor; and
- (25) FIG. 24 is a side view of the needle assembly of FIG. 23.

GENERAL DESCRIPTION

(26) A drug delivery device, in the form of an injector, may include various systems for limiting the delivery of a medical fluid or drug product in case of movement of (e.g., removal of) the injector relative to a patient as determined by a proximity sensor. The drug delivery system may in the alternative or in addition include systems for indicating the amount of medical fluid or drug product delivered (or not delivered) in the case of movement of (e.g., removal of) the injector relative to the patient as determined by the proximity sensor. The injector may be, for example, an on-body injector or a hand-held autoinjector. The injector may be used with one of a variety of medical fluids or drug products.

(27) In one embodiment, an injector comprises a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends. The injector further comprises a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir, and a spring coupled to the plunger assembly to move the plunger between the first and second ends. The injector also comprises a lock selectively coupled to one of the plunger assembly and the spring to limit movement of the plunger between the first and second ends with the lock coupled to the one of the plunger assembly and the spring, a housing, the reservoir, spring, and lock disposed within the housing, and a proximity sensor coupled to the lock and moveable relative to the housing, the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the lock coupled to the one of the plunger assembly and the spring with the proximity sensor in the first sensor state.

(28) According to another embodiment, the plunger assembly includes a plunger arm attached to the plunger, the lock has a wall that abuts the plunger arm to limit movement of the plunger when the lock is coupled to the plunger assembly, and the proximity sensor is coupled to the wall, where the wall abuts the plunger arm with the proximity sensor in the first sensor state and the wall is spaced from the plunger arm with the proximity sensor in the second sensor state.

(29) According to another embodiment, the plunger arm has at least one shoulder formed thereon, and the wall abuts the at least one shoulder of the plunger arm to limit movement of the plunger when the lock is coupled to the plunger assembly.

(30) According to another embodiment, the injector further comprises a plate disposed at least partially within the housing and coupled to the housing to translate relative to the plunger arm, the plate having a first end that defines the wall of the lock and a second end that defines the proximity sensor.

(31) According to another embodiment, the injector further comprises a lever having a first end disposed within the housing and defining the wall of the lock, and a second end disposable outside the housing and defining the proximity sensor.

(32) According to another embodiment, the lock has a wall that abuts the spring to limit movement of the plunger when the lock is coupled to the spring, and the proximity sensor is attached to the wall, where the wall abuts the spring with the proximity sensor in the first sensor state and the wall is spaced from the spring with the proximity sensor in the second sensor state.

(33) According to another embodiment, the injector further comprises a plate disposed at least partially within the housing and coupled to the housing to translate relative to the spring, the plate

having a first end that defines the wall of the lock and a second end that defines the proximity sensor.

(34) According to another embodiment, the injector further comprises a lever having a first end disposed within the housing and defining the wall of the lock, and a second end disposable outside the housing and defining the proximity sensor.

(35) According to another embodiment, the injector further comprises a spring coupled to the proximity sensor, the spring biasing the proximity sensor toward the first sensor state.

(36) According to another embodiment, the lock is reversibly coupled to the one of the plunger assembly and the spring.

(37) According to another embodiment, the lock is irreversibly coupled to the one of the plunger assembly and the spring.

(38) According to another embodiment, the proximity sensor has a third sensor state wherein the proximity sensor is retracted toward the housing, the lock being coupled to the one of the plunger assembly and the spring with the proximity sensor in the third sensor state, the lock being reversibly coupled to the one of the plunger assembly and the spring with the sensor occupying the third sensor state subsequent to occupying the second sensor state, and the lock being irreversibly coupled to the one of the plunger assembly and the spring with the sensor occupying the first sensor state subsequent to occupying one of the second and third sensor states.

(39) According to another embodiment, the lock prevents movement of the plunger between first and second ends with the lock coupled to the one of the plunger assembly and the spring.

(40) According to another embodiment, the lock limits movement of the plunger between first and second ends with the lock coupled to the one of the plunger assembly and the spring.

(41) According to another embodiment, the injector is a hand-held autoinjector, the injector comprising a needle shield that defines the proximity sensor.

(42) According to another embodiment, the injector is an on-body injector, the injector comprising a surface having adhesive applied thereto to attach the injector to a body of a patient.

(43) According to a further embodiment, an injector comprises a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends, a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir, and a gas source having an operational state wherein the gas source is in fluid communication with the plunger to move the plunger between the first and second ends. The injector further comprises a lock comprising a vent selectively coupled to the gas source to limit movement of the plunger between the first and second ends, a housing, the reservoir, gas source, and lock disposed within the housing, and a proximity sensor coupled to the lock and moveable relative to the housing, the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the vent coupled to the gas source with the proximity sensor in the first sensor state.

(44) According to another embodiment, the gas source is a pressurized container of gas.

(45) According to another embodiment, the gas source is a container of a material capable of a phase change from liquid to gas or solid to gas.

(46) According to another embodiment, the vent comprises a seal.

(47) According to another embodiment, the vent comprises a piercable septum.

(48) According to a still further embodiment, an injector comprises a reservoir including a bore having a first end and a second end, and a plunger assembly including a plunger moveable within the bore between the first and second ends, a cannula having an operational state wherein the cannula is connected in fluid communication with the reservoir, and a spring coupled to the plunger assembly to move the plunger between the first and second ends. The injector further comprises an indicator selectively mechanically coupled to the plunger assembly, a housing, the reservoir, spring, and indicator at least partially disposed within the housing, and a proximity sensor coupled to the

indicator and moveable relative to the housing, the proximity sensor having a first sensor state wherein the proximity sensor extends fully from the housing and a second sensor state wherein the proximity sensor is retracted toward the housing, the indicator coupled to the plunger assembly with the proximity sensor in the second sensor state and the indicator decoupled from the plunger assembly with the proximity sensor in the first sensor state.

(49) According to another embodiment, the plunger assembly comprises a plunger arm attached to the plunger, the indicator being mechanically coupled to the plunger arm through a gear train having at least one gear that is moveable into and out of engagement, and the proximity sensor is coupled to the at least one gear to move the at least one gear out of engagement and decouple the indicator from the plunger arm with the proximity sensor in the first sensor state.

DETAILED DESCRIPTION

(50) Turning now to FIGS. 1-3, a drug delivery device such as an injector **100** includes a reservoir **102**, a plunger assembly **110**, a cannula **114**, and a drive **116**. The reservoir **102** includes a bore **104** having a first end **106** and a second end **108**. The plunger assembly **110** includes a plunger **112** moveable within the bore **104** of the reservoir **102** between the first and second ends **106**, **108**. The cannula **114** includes an operational state wherein the cannula **114** is connected in fluid communication with the reservoir **102**. The drive **116**, in the form of a spring, is coupled to the plunger assembly **110** to move the plunger **112** between the first and second ends **106**, **108**.

(51) According to the embodiments illustrated in FIGS. 1-12, the injector **100** is also provided with a lock **118** selectively coupled to one of the plunger assembly **110** and the drive **116** to limit movement of the plunger **112** between the first and second ends **106**, **108** of the bore **104**. For example, the lock **118** may be coupled to the one of the plunger assembly **110** and the spring **116**. See, e.g., FIGS. 1-10. Another embodiment of the lock **118** is illustrated in FIGS. 11 and 12.

(52) The injector **100** includes a proximity sensor **120** coupled to the lock **118** and moveable relative to a housing **122** in which the reservoir **102**, spring **116**, and lock **118** are disposed. The proximity sensor **120** has a first sensor state (or position) wherein the proximity sensor **120** extends (e.g., extends fully) from the housing **122** (see, e.g., FIGS. 1 and 2) and a second sensor state (or position) wherein the proximity sensor **120** is retracted toward and into the housing **122** relative to the first sensor state (see, e.g., FIG. 3). The lock **118** is coupled to the one of the plunger assembly **110** and the spring **116** with the proximity sensor **120** in the first sensor state so as to limit or prevent movement of the plunger **112**.

(53) In certain embodiments, the reservoir **102** may be in the form of a pre-filled syringe, in which case the cannula **114** may be in the form of a needle fixedly or securely attached to a hub **130** at the second end **108** of the reservoir **102**. See, FIG. 1. According to other embodiments, the reservoir **102** may be pre-filled, but the cannula **114** may be moveable relative to the hub **130** of the reservoir **102**. In some embodiments, the hub **130** can include a septum such that the cannula **114** can be moved in the direction of the hub **130** to pierce the septum in the operational state. In certain embodiments, an assembly may be disposed between the cannula **114** and the hub **130** to dispose the cannula **114** into fluid connection with the hub **130** in the operational state, which assembly may also insert the cannula **114** into the patient. In some embodiments, a barrier may be disposed about the end of the cannula **114** that pierces the septum to preserve the sterility at the junction between the cannula **114** and the septum. In some embodiments, the cannula **114** either may be fixedly or securely attached or may be moveable relative to the reservoir **102** (more particularly, the hub **130**), but the reservoir **102** may not be pre-filled.

(54) As illustrated in FIG. 1, the plunger assembly **110** may include a plunger arm **140** attached to the plunger **112**. The lock **118** has a wall **142** that abuts the plunger arm **140** to limit movement of the plunger **112** when the lock **118** is coupled to the plunger assembly **110**. The proximity sensor **120** is coupled to the wall **142** (as illustrated, the sensor **120** is integral, or one piece, with the wall **142**), such that the wall **142** abuts the plunger arm **140** with the proximity sensor **120** in the first sensor state (FIGS. 1 and 2) and the wall **142** is spaced from the plunger arm **140** with the

proximity sensor **120** in the second sensor state (FIG. 3).

(55) In some embodiments, the plunger arm **140** may have at least one shoulder **144** (FIG. 1) formed thereon, and the wall **142** abuts the at least one shoulder **144** of the plunger arm **140** to limit and/or prevent movement of the plunger **112** when the lock **118** is coupled to the plunger assembly **110**. As illustrated in FIG. 1, the plunger arm **140** has a section of its length (i.e., a dimension of the plunger arm **140** extending in a direction along a longitudinal axis **148** of the plunger arm **140**) that has at least one feature **150** that defines the at least one shoulder **144**. For example, the plunger arm **140** may include a shaft **152** to which is attached one or more features **150** that include protrusions **154** (e.g., teeth). In some embodiments, the protrusions **154** can be formed integrally (as one piece) with the shaft **152**. Spaces or notches **156** between adjacent protrusions **154** permit the wall **142** to be disposed between adjacent protrusions **154**, with the protrusion **154** positioned furthest longitudinally from the plunger **112** defining the shoulder **144** along a surface of the protrusion **154** that extends perpendicularly to the axis **148**. It will be recognized that the features **150** of the embodiment illustrated in FIGS. 1-3 may also be described as one or more spaces or notches **156** formed in the plunger arm **140**, the spaces or notches **156** being bounded by the material of the plunger arm **140** previously described as shaft **152** and protrusions **154**. These are equivalent descriptions for the disclosed subject matter.

(56) As noted above, the lock **118** includes a wall **142** that fits in the spaces or notches **156** thereby coupling the lock **118** to the plunger assembly **110**. According to the embodiment illustrated in FIGS. 1-3, the injector **100** includes a plate **158** disposed at least partially within the housing **122** and coupled to the housing **122** to translate relative to the plunger arm **140**. The plate **158** may have a first end **160** that defines the wall **142** of the lock **118** and a second end **162** that defines the proximity sensor **120**. See FIGS. 2 and 3. While the wall **142** has been described as being part of the plate **158**, which is depicted in FIGS. 2 and 3, for example, as a generally planar structure, this is merely an example. The wall **142** of the lock **118** could also be defined by part of a pin, a fork, or any other component not expressly depicted but capable of serving the intended purpose.

(57) In some embodiments, the injector **100** may include a spring **166** coupled to the proximity sensor **120** (via the plate **158**), the spring **166** biasing the proximity sensor **120** (again, via the plate **158**) toward the first sensor state. In certain embodiments, the second end **160** may include a guide **172** about which the spring **166** is disposed, a first end **174** of the spring **166** abutting a shoulder **176** of the plate **158** and a second end **178** abutting an interior surface **180** of the housing **122**. While a coil spring **166** is illustrated, any of a number of different biasing elements may be used, which biasing elements may bias the first end **162** of the plate **158** outwardly from the housing **122**.

(58) In some embodiments, the injector **100** may include guides attached to the housing **122** that prevent the plate **158** from other than translational movement along a line of motion **182**. For example, a guide may be disposed on either side of the plate (i.e., to the right or the left of the plate **158** as illustrated in FIG. 2) to prevent lateral movement relative to the line of motion **182**.

(59) Disposed between the first and second ends **160**, **162** of the plate **158** is an aperture **184**. While the aperture **184** appears to be closer to the second end **160** of the plate **158** than the first end **162**, other arrangements are possible. Furthermore, while the aperture **184** appears to be fully surrounded by the plate **158**, such that the entire circumference of the aperture **184** is defined by the plate **158**, it is also possible that at least a section of the aperture **184** is not surrounded by the plate **158** and thus the plate **158** only partially defines the circumference of the aperture **184**.

Furthermore, while the circumference of the aperture **184** appears oblong or keyhole in shape in FIG. 2, the aperture **184** is not limited to any particular shape.

(60) As illustrated in FIGS. 1 and 2, when the injector **100** is not disposed on the surface of the patient's skin, the proximity sensor **120** extends from the housing **122** as a consequence of the force applied to the plate **158** by the spring **166**. See FIG. 2. In turn, the plate **158** is positioned relative to the plunger assembly **110**, and in particular the plunger arm **140**, such that the plate **158** resides within one of the spaces or notches **156**. Compare FIGS. 1 and 2. When the injector **100** is disposed

on the surface of the patient's skin, the proximity sensor **120** is moved into the housing **122** against the bias of the spring **166**. As a consequence, the plate **158** is moved into a position where the aperture **184** is aligned with the plunger arm **140**, such that the plate **158** no longer resides within one of the spaces or notches **156**. See FIG. **3**. This permits movement of the plunger arm **140** and associated plunger **112** as a consequence of the force applied to the plunger arm **140** by the spring **116**.

(61) During the motion of the plunger arm **140** toward the right relative to the orientation of FIG. **1**, the injector **100** may become detached or displaced from the patient's skin. In such a case, the plate **158** would be permitted to move under bias of the spring **166** such that the aperture **184** is no longer aligned with the plunger arm **140**, and instead the plate **158** becomes disposed within one of the spaces or notches **156**. In this version, the spring **166** transmits information to the lock **118** in the form of a mechanical force indicative of the position of the plate **158** and proximity sensor **120**. This force can cause engagement between the plate **158** and one of the protrusions **154** that would prevent further motion of the plunger arm **140** at the urging of the spring **116**, and would limit the amount of medical fluid or drug product ejected from the reservoir **102**. That is, according to certain embodiments, engagement between the plate **158** and a protrusion may prevent any further medical fluid or drug product from passing through and out of the cannula **114**. According to other embodiments, the plunger arm **140** and associated plunger **112** may travel some distance after the plate **158** becomes disposed within a space or notch **156** but before the plate **158** engages a protrusion **154**, such that a limited amount of medical fluid or drug product may pass out of the reservoir through the cannula **114** even after activation of the lock **118**. It will be recognized that by limiting the amount of medical fluid or drug product ejected from the reservoir **102**, while arresting the overall motion of the plunger **112**, significant advantages may still be obtained.

(62) It is not necessary that the lock **118** and proximity sensor **120** be defined by a translating plate, such as is illustrated in FIGS. **1-3**. The embodiment illustrated in FIGS. **4** and **5** includes a different structure that defines both a lock **118** and a proximity sensor **120**. Because many of the structures of the embodiment of the injector illustrated in FIGS. **4** and **5** are identical to those of the embodiment of FIGS. **1-3**, like elements have like reference numerals.

(63) The structure of the embodiment illustrated in FIGS. **4** and **5** that defines both the lock **118** and the proximity sensor **120** is a lever **190**. As shown in FIG. **5**, the lever **190** has a first end **192** disposed within the housing **122** and defining the wall **142** of the lock **118**, and a second end **194** disposable outside the housing **122** and defining the proximity sensor **120**. Disposed between the first and second ends **192**, **194** is a pivot **196** that may be attached to the housing **122**.

(64) The injector **100** may include a spring **198** coupled to the proximity sensor **120**, the spring **198** biasing the proximity sensor **120** (lever **190**) toward the first sensor state with the sensor **120** extending from the housing **120**. As illustrated, the spring **198** is a torsion spring that is disposed at the pivot **196** and that applies a force to the lever **190** to one or the other side of the lever **190** to bias the lever **190**, and thus the sensor **120**, toward the first sensor state (i.e., in a clockwise direction relative to the orientation of FIG. **5**). It will be recognized that other biasing elements may be used instead. For example, a compression spring may be disposed between the first end **192** of the lever **190** and the housing **122** to urge the lever **190**, and thus the sensor **120**, toward the first sensor state.

(65) Some change in orientation of the elements of the injector **100** may be required to utilize the lever **190**. Because the motion of the second end **194** of the lever **190** into the housing **122** causes the first end **192** to move counter-clockwise, the orientation of the plunger arm **140**, and the spaces or notches **156** on the plunger arm **140**, may be the reverse of that illustrated in the embodiment of FIGS. **1-3**. The spaces or notches **156** may be positioned below the plunger arm **140**, as depicted in FIG. **4**. In this fashion, movement of the lever **190** caused by the motion of the second end **194** upon application of the injector **100** to the patient's skin would cause the first end **192** to move away from the plunger arm **140** and out of its respective space or notch **156**. And upon removal of

the injector **100** from the patient's skin, whether the spring **198** would cause the first end **192** of the lever **190** to move toward the plunger arm **140** and into a respective space or notch **156** to lock the device. In this version, the spring **198** transmits information to the lock **118** in the form of a mechanical force indicative of the position of the lever **190** and thus the proximity sensor **120**.

(66) One advantage of the lever **190** is that it can permit a magnification of the travel of the second end **194**, which in turn can increase the sensitivity of the proximity sensor **120**. In particular, as illustrated in FIG. **6**, the second end **194** moves a distance d relative to the surface of the housing **122** (for purposes of this discussion, this surface is also the patient contact surface). The first end **192** in turn moves a distance D relative to the same surface of the housing **122**. These distances (d , D) are related to a length L of the lever **190**, a length 1 of the lever **190** to the second end **194** and the offset H of the pivot **196** relative to the surface of the housing **122**. If the length 1 is one-third of the length L , then changes in the distance d will result in changes approximately twice as large for the distance D (assuming a relatively small offset H). As long as length 1 is less than half of length L , an increased sensitivity to motion will be realized.

(67) The sensitivity of the embodiment utilizing a lever **190** may also be influenced by changes in the shape of the lever **190**. The lever **190** illustrated in FIGS. **4-6** is relatively straight.

Alternatively, the lever **190** may be shaped with a bend between the two ends **192**, **194** to change the relationship between the movement of the second end **194** to the first end **192**.

(68) While the embodiments of FIGS. **1-3** and **4-6** have been illustrated with a single structure (plate **158**, lever **190**) defining both the lock **118** and the proximity sensor **120**, other embodiments may not be so limited. For example, both the lock **118** and the sensor **120** may each be defined by a translating plate, similar to the plate **158**, except that the first plate that defines the lock **118** may be disposed in a first region of the injector **100** and the second plate that defines the sensor **120** may be disposed in a second region of the injector **100** spatially removed from the first region. The second plate that defines the sensor **120** may be coupled to the first plate that defines the lock **118** by a lever or other mechanical system disposed within the housing. A pivot for the lever coupling the sensor **120** to the lock **118** may be selected to provide greater sensitivity that may be possible if a single structure defined both lock **118** and sensor **120**. In addition, a lever may permit the sensor **120** to be disposed closer to the cannula **114** while the lock **118** is disposed proximate to the plunger arm **140**, because the lock **118** and the sensor **120** are not defined by a single structure.

(69) In further embodiments, other structures or assemblies of structures may be used to transform motion of the sensor **120** into motion of the lock **118**. The foregoing example is simply one possible embodiment of a class of embodiments, individual examples of which may permit greater sensitivity while not necessarily permitting the lock **118** and the sensor **120** to be disposed in spatially remote sections of the injector, and vice versa. Further, embodiments may provide other functions in addition or in substitution for those previously mentioned. For example, an intermediate structure or assembly may couple a sensor **120** moving in a first direction to a lock **118** moving in a second direction, which direction may be different from the first (i.e., the sensor **120** may travel along a first line of motion while the lock **118** moves along a second line of motion that is at right angles to the first line of motion).

(70) As noted above, the lock **118** may be coupled to one of the plunger assembly **110** and the spring **116**. As illustrated in FIGS. **7-10**, the lock **118** may be coupled to the spring **116** instead of the plunger assembly **110**.

(71) According to this embodiment, the plunger assembly **110** may or may not include a plunger arm **140**. As illustrated in FIGS. **7-10**, the plunger assembly **110** does not include a plunger arm **140**, but the spring **116** acts directly against the plunger **112**, the force applied by the spring **116** causing motion of the plunger **112**. It will be recognized that the embodiments of the lock **118** described below would work equally as well if the spring **116** applied its force to an intermediate structure, such as a plunger arm **140**, instead of directly to the plunger **112**.

(72) According to the embodiment illustrated in FIG. 7, the lock **118** has a wall **200** that abuts the spring **116** to limit movement of the plunger **112** when the lock **118** is coupled to the spring **116**. In some embodiments, as illustrated, the spring **116** may be a coil spring, and the wall **200** may be disposed between adjacent coils **202** of the spring **116** to limit the force applied to the plunger **112**. The proximity sensor **120** is attached to the wall **200**, and the wall **200** abuts the spring **116** with the proximity sensor **120** in the first sensor state and the wall **200** is spaced from the spring **116** with the proximity sensor **120** in the second sensor state. In some embodiments, the injector may include a spring **204** coupled to the proximity sensor **120**, the spring **204** biasing the proximity sensor **120** toward the first sensor state.

(73) The embodiment illustrated in FIGS. 7 and 8 may include a plate **206** that defines the wall **200**. The plate **206** may be disposed at least partially within the housing **122** and coupled to the housing **122** to translate relative to the spring **116**. The plate **206** may have a first end **208** that defines the wall **200** of the lock **118** and a second end **210** that defines the proximity sensor **120**. In some embodiments, the plate **206** includes an aperture **212** that is sized such that the spring **116** may pass through the plate **206** when the sensor **120** is in the second sensor state, the consequence of which would be for the spring **116** to move freely without interference from the plate **206** and exert a force against the plunger **112**.

(74) The discussion relative to the operation and configuration of the plate **158** relative to the plunger arm **140** applies with equal force relative to the operation and configuration of the plate **206** relative to the spring **116**. For example, like the version in FIGS. 1-3, when the injector **100** in FIGS. 7 and 8 is removed from the patient, the spring **204** transmits information to the plate **206** of the lock **118** in the form of a mechanical force indicative of the position of the proximity sensor **120**.

(75) In some embodiments, the injector may include a lever **220**, as illustrated in FIGS. 9 and 10, in substitution for the plate **206**. As shown in FIG. 10, the lever **220** may have a first end **222** disposed within the housing **122** and defining the wall **200** of the lock **118**. The lever **220** may also have a second end **224** disposable outside the housing **122** and defining the proximity sensor **120**. This injector may also include a spring **228** coupled to the proximity sensor **120**, the spring **228** biasing the proximity sensor **120** (lever **220**) toward the first sensor state. For example, the spring **228** may be a torsion spring.

(76) The discussion relative to the operation and configuration of the lever **190** relative to the plunger arm **140** applies with equal force relative to the operation and configuration of the lever **220** relative to the spring **116**. For example, like the version in FIGS. 4-6, when the injector **100** in FIGS. 9 and 10 is removed from the patient, the spring **228** transmits information to the wall **200** of the lock **118** in the form of a mechanical force indicative of the position of the lever **220** and proximity sensor **120**. In addition, the discussion above relative to the lock **118** and the sensor **120** being defined by a single structure or by separate structures coupled by one or more intermediate structures or assemblies of structures applies with equal force regarding the embodiments illustrated in FIGS. 7-10 as it did relative to the embodiments illustrated in FIGS. 1-6.

(77) According to some embodiments, the lock **118** may be reversibly coupled to the one of the plunger assembly **110** and the spring **116**. For example, if the injector **100** is removed from the patient's skin during administration, the lock **118** would limit or prevent further administration of the medical fluid or drug product. But, in embodiments where the lock **118** is reversible, the lock **118** may be unlocked, or decoupled, if the proximity sensor **120** determines that the injector **100** has been replaced against the patient's skin. In other embodiments, the lock **118** may be irreversibly coupled to the one of the plunger assembly **110** and the spring **116**. Such embodiments would prevent the injector **100** from operating even if the injector **100** was re-positioned against the patient's skin. In some embodiments, the decision to make the lock **118** reversible or irreversible may be selected by the patient or caregiver using the delivery device.

(78) One example of an irreversible lock **118** is illustrated in FIG. 1, wherein all but one of the

features **150** extending up from the plunger arm **140** can include transverse teeth **151** extending into spaces or notches **156** between the features **150**. In the version of FIG. **1**, all of the features **150** include a tooth **151** except for the feature **150** located in closest proximity to the plunger **112**. So configured, before the injector **100** is ever operated, the lock **118** resides in the space or notch **156** disposed between the plunger **112** and the adjacent toothless feature **150**. Upon the injector **100** being positioned against a patient's skin, the proximity sensor **120** and lock **118** move into the injector **100** to allow the plunger assembly **110** to drive the plunger **112** and administer drug to the patient. This causes the plunger arm **140** to move to the right relative to the orientation of FIG. **1**. Then, once the injector **100** is removed from the patient's skin, the spring **166** biases the lock **118** and proximity sensor **120** back out of the injector **100** and into the position illustrated in FIG. **1**, for example. Here, the lock **118** would be positioned in a space or notch **156** between two features **150**, one of which includes a tooth **151**. The tooth **151** resides above the lock **118**, or in a recess formed in the lock **118**, for example, such that a bottom surface of the tooth **151** engages an upward facing surface of the lock **118**, thereby preventing the lock **118** and proximity sensor **120** from moving back up into the injector **100**. While FIG. **1** illustrates the features **150** as having the teeth **151**, this would only apply to those injectors **100** where the lock **118** is irreversible. For a reversible lock **118**, the injector **100** would not include the teeth **151**.

(79) Some embodiments of the injector **100** may have a lock **118** that is reversibly coupled and irreversibly coupled to one of the plunger assembly **110** and the spring **116** according to the state of the proximity sensor **120**. For example, the proximity sensor **120** may have a third sensor state wherein the proximity sensor **120** is retracted toward the housing relative to the first sensor state but not as far as the second sensor state. In some embodiments, the third sensor state can be considered an intermediate state between the first and second sensor states. With the proximity sensor **120** in the third sensor state, the lock **118** can be coupled to the one of the plunger assembly **110** and the spring **116** to limit or prevent further movement of the plunger **112**. Further, the lock **118** may be reversibly coupled to the one of the plunger assembly **110** and the spring **116** with the proximity sensor **120** in the third sensor state subsequent to occupying the second sensor state. And, the lock **118** can be irreversibly coupled to the one of the plunger assembly **110** and the spring **116** with the proximity sensor **120** in the first sensor state subsequent to occupying one of the second and third sensor states.

(80) To illustrate the foregoing, consider the following example. A patient applies the device to his or her skin. As a consequence, the proximity sensor **120** is disposed wholly or partially in the housing **120**. As a further consequence, the lock **118** is disengaged or uncoupled from the plunger assembly **110** or the spring **116**, and the device begins injecting the medical fluid or drug product into the patient.

(81) At a time thereafter, but before the full dose has been administered, the delivery device may be disturbed, causing a separation of the device from the patient's skin, as determined by the proximity sensor **120**. This disturbance may cause the cannula **114** to only shallowly penetrate the patient's skin, but may not separate the injector **100** from the patient's skin to such a degree that the cannula **114** is completely removed or the sterility of the access site is compromised. In such a circumstance, the delivery of the medical fluid or drug product may be interrupted by the coupling of the lock **118** with the plunger assembly **110** or the spring **116** to prevent medical fluid or drug product leakage that may occur because of shallow cannula **114** penetration. But, this coupling is reversible so that an application of force that moves the cannula **114** back to its proper depth, as determined by the proximity sensor **120**, will cause the lock **118** to decouple from the plunger assembly **110** or spring **116**, thereby re-starting administration of the medical fluid or drug product.

(82) By having a reversible coupling, the medical fluid or drug product remaining in the reservoir **102** can be successfully delivered and would not end up being treated as waste. Consequently, reversible coupling of the lock **118** to the plunger assembly **110** or the spring **116** would provide an opportunity to save a dose in the event of a minor disturbance or use error. Additionally, this may

provide a higher success rate on challenging anatomies by allowing applied pressure to fix or correct an “error” state and permit delivery to resume.

(83) If, however, the delivery device becomes disturbed to such a degree that the cannula **114** is completely removed from the patient (for example, the device falls completely off the patient) before the full dose has been administered, then it may be undesirable to permit the patient or caregiver to reapply the injector **100** to the patient. Consequently, to reduce the likelihood that the patient or caregiver would attempt to reapply the injector **100**, the lock **118** may irreversibly couple to the plunger assembly **110** or spring **116** to prevent the injector **100** from any further drug delivery.

(84) To illustrate this concept, consider an embodiment where insertion of the cannula **114** to a depth of 6 mm is preferred and a depth of 3 mm does not cause significant leakage—the 3 mm depth may be referred to as a delivery continuation threshold, and may represent the point, degree or level under which continued delivery presents an unacceptable risk of leakage or improper administration. When the injector **100** is applied and the proximity sensor **120** is in its fully retracted state, this may correspond to a state for the cannula **114** where the cannula **114** is inserted to a depth of 6 mm. If the injector **100** is disturbed, and the sensor **120** moves to a state where it is not fully retracted or fully extended corresponding to a cannula depth of 3 mm (e.g., the sensor **120** extends 3 mm from the housing **122**), then the injector **100** may permit continued injection of the medical fluid or drug product. However, if the sensor **120** moves to a state corresponding to cannula depth of 2 mm (e.g., the sensor **120** extends 4 mm from the housing **122**), then the injector **100** may reversibly limit injection of the medical fluid or drug product by reversibly coupling the lock **118** to one of the plunger assembly **110** and the spring **116**. If the sensor **120** then moves back to a state corresponding to a cannula depth of 3 mm (e.g., the sensor **120** extends 3 mm from the housing **122**) or more, then the injection may resume.

(85) However, if the sensor **120** moves to a state corresponding to a cannula depth of 0 mm (e.g., the sensor **120** extends 6 mm from the housing **122**), then the injection would be permanently and irreversibly limited through the irreversible coupling of the lock **118** to the plunger assembly **110** or spring **116**. A depth of 0 mm may be referred to as a device removal threshold, and may represent the point, degree or level at which the device has likely been compromised, removed or the cannula **114** disturbed such that further delivery/reapplication is undesirable or unsafe. A similar outcome may result where the cannula depth is not 0 mm, but some non-zero value (e.g., 1 mm). A non-zero value may be selected because even if the cannula **114** does not come completely out of the patient's skin, this may still be considered to present an unacceptable risk. Alternatively, a non-zero value may be selected so that it is not a requirement that the sensor **120** determine complete separation to fail-safe in case of a false negative for complete separation (e.g., partial interference between the housing **122** and the sensor **120**).

(86) The specifics of any particular embodiment may be influenced by, for example, the distance to which the proximity sensor **120** extends from the housing **120**, the length of the cannula **114**, the desired depth of the insertion of the cannula **114** into the patient, and the relationship between the depth of cannula insertion to the degree of leakage from the access site.

(87) According to any of the foregoing embodiments, the lock **118** may limit movement of the plunger **112** between first and second ends **106**, **108** so as to completely prevent movement of the plunger **112** between the first and second ends **106**, **108** upon the lock **118** being coupled to the one of the plunger assembly **110** and the spring **116**. Alternatively, according to any of the foregoing embodiments, the lock **118** may only limit movement of the plunger **112** between first and second ends **106**, **108** without completely preventing movement of the plunger **112** between the first and second ends **106**, **108** upon the lock **118** being coupled to the one of the plunger assembly **110** and the spring **116**. That is, it is not a requirement of all embodiments that the engagement or coupling of the lock **118** immediately result in the complete cessation of movement of the plunger **112**. In certain circumstances, the plunger **112** may continue its movement over a limited length of travel

after activating the lock **118**. The specific distance that the plunger **112** may travel may be known and predetermined, as a consequence of the structure and operation of the lock **118**, or the distance may be approximated based on the structure and operation of the lock **118** or the tolerances in the manufacture of the injector **100**. In some embodiments, any limited length of travel of the plunger **112** that is allowed after engagement or coupling of the lock **118** can occur, for example, because of the geometry of the wall **142** that forms the lock **118** and the spaces or notches **156** on the plunger arm **140**. That is, in some embodiments, the spaces or notches **156** may have a dimension that is greater than a thickness or width of the wall **142**. As such, it is foreseeable that in some embodiments, the plunger **112** can continue to move to the extent that the spaces or notches **156** are wider, even after the lock **118** has been engaged or coupled.

(88) Those embodiments described above wherein the lock **118** engages or couples to the plunger assembly **110**, and in particular the plunger arm **140**, may be used with embodiments of an injector **100** wherein the drive **116** is not in the form of a spring. For example, any device that acts on the plunger arm **140** may be used as a drive **116** in the circumstance wherein the lock **118** engages or couples to the plunger assembly **110** to limit the injection of medical fluid or drug product from the injector **100**. Where the drive **116** is not in the form of a spring, other considerations may be required when the injection is limited by interaction between the lock **118** and the drive **116**.

(89) According to another group of embodiments illustrated in FIGS. **11** and **12**, the injector includes a reservoir **102** including a bore **104** having a first end **106** and a second end **108**, and a plunger assembly **110** including a plunger **112** moveable within the bore **104** between the first and second ends **106**, **108**. The injector **100** also includes a cannula **114** having an operational state wherein the cannula **114** is connected in fluid communication with the reservoir **102**.

(90) Embodiments of the injector **100** illustrated in FIGS. **11** and **12** include a drive **116** in the form of a gas source **250** having an operational state wherein the gas source is in fluid communication with the plunger **112** to move the plunger **112** between the first and second ends **106**, **108**. The gas source **250** may be a container of pressurized gas. In some embodiments, the gas source **250** may be a container of a material capable of a phase change from liquid to gas or solid to gas. When activated, the gas source **250** applies positive pressure on the plunger **112** of the plunger assembly **110**. In some embodiments, the gas source **250** can include a frangible seal (similar to that on a convention CO.sub.2 cartridge) that is punctured by a movable needle, for example, upon activation of the injector **100**. The movable needle may be operably connected to the proximity sensor **120** such that movement of the proximity sensor **120** automatically moves the needle to puncture the frangible seal and activate the gas source **250**. In other embodiments, the injector **100** may include a separate button operably connected to the movable needle such that a user of the injector **100** must depress the button to move the needle and break the frangible seal on the gas source **250**. In still other embodiments, the injector **100** can include a mechanical or electro-mechanical valve that opens and activates the gas source **250** upon activation of the injector **100**, or some other structure capable of achieving the intended objective.

(91) The injector **100** according to such an embodiment includes a lock **118** comprising a vent **252** selectively coupled to the gas source **250** to limit or prevent movement of the plunger **112** between the first and second ends **106**, **108**. When the vent **252** is opened, positive pressure from the gas source **250** can exhaust through the vent **252** instead of applying a force against the plunger **112** of the plunger assembly **110**. The vent **252** could be provided by a number of different mechanisms, and may be dependent upon the manner in which the positive pressure is delivered to the plunger **112**. For example, the vent **252** could be defined by a rupturable wall or a moveable seal, which could provide an irreversible or a reversible lock. According to one embodiment, the vent **252** may be in the form of a duck-bill valve.

(92) Additionally, when the gas source **250** and vent **252** have a frangible seal or rupturable wall, the injector **100** could be considered as having an “irreversible” lock. For example, when the gas source **250** and vent **252** are opened, the vent **252** limits or prevents the positive gas pressure from

being applied against the plunger assembly **110**. In such an assembly, the open “vent” causes the lock to operate irreversibly because there is only a single pressure source (i.e., the gas source **250**), and once its seal is broken, its contents will be expelled until it is empty, especially if a seal of the vent **252** is also broken. Therefore, if the injector **100** is removed from the patient prematurely, the gas source **250** continues to expel gas which is then exhausted out of the vent **252**. Any subsequent attempt to re-administer will fail if the gas source **250** is spent.

(93) In some embodiments, the vent **252** may include a seal **254**, such as a piercable septum or a duck-bill valve, and a vent cannula **256**. The vent cannula **256** pierces the seal **254** to provide an exhaust pathway out of the vent **252**. The reservoir **102**, gas source **250**, and lock **118** may be disposed within the housing **122**, while the proximity sensor **120** may be disposed in whole or in part outside of the housing **122**.

(94) According to some embodiments, the proximity sensor **120** is coupled to the lock **118** and moveable relative to the housing **122**. The proximity sensor **120** has a first sensor state wherein the proximity sensor **120** extends (e.g., extends fully) from the housing **122** and a second sensor state wherein the proximity sensor **120** is retracted toward and into the housing **122** relative to the first sensor state. The vent **252** is coupled to the gas source **250** with the proximity sensor **120** in the first sensor state.

(95) The injector **100** illustrated in FIGS. **11** and **12** includes a chamber **258** that connects the gas source **250** with the reservoir **102**. The vent **252** is disposed in a wall of the chamber **258**. The vent cannula **256**, which may be in the form of a blunt, open-ended needle, is attached to a first end **260** of a vent lever **262**. The second end **264** of the vent lever **262** is coupled to the proximity sensor **120**, which may be in the form of cup-shaped button **266** that extends through an aperture **268** in a wall of the housing **122**. Movement of the button **266** is transmitted by the vent lever **262** to the vent cannula **256**.

(96) The proximity sensor **120**, and in particular the button **266**, is biased outwardly from the housing **122** by a spring **274**. The spring **274** may be disposed between a support **276** and the second end **264** of the vent lever **262**, the button **266** being disposed opposite the support **276** relative to the end **264** of the vent lever **262**.

(97) In operation, with the proximity sensor **120** extended (see FIG. **12**) and the gas source **250** not yet activated, the vent cannula **256** engages the seal **254** to exhaust the chamber **258**. Because the gas source **250** has not yet been activated, there is no effect on the movement of the plunger **112**. However, once the proximity sensor **120** retracts into the housing **122**, the vent cannula **256** is moved so that the vent cannula **256** no longer interrupts the seal **254** and a positive pressure may be generated in the chamber **258** by the gas source **250**. If the proximity sensor **120** extends before the full dose is administered, such as by removal of the injector **100** from the patient, the spring **274** biases the vent lever **262** such that the vent cannula **256** is moved into engagement with the seal **254**, and the positive pressure in the chamber **258** is exhausted through the vent cannula **256**. This decreases the pressure in the chamber **258**, causing the movement of the plunger **112** to be limited, or even stopped. In this configuration, the spring **274** transmits information to the lock **118**, and specifically the vent lever **262** and vent cannula **256**, in the form of a mechanical force indicative of the position of the vent lever **262** and the proximity sensor **120**.

(98) The embodiment provided in FIGS. **11** and **12** is intended to be an example only. Other structures may be substituted for the seal **254**, vent cannula **256**, and vent lever **262**. While a chamber **258** has been provided in communication with the reservoir **102** and the gas source **250**, the chamber **258** may not be present in other embodiments. Additionally, in some embodiments, the cannula **114** may be an open ended needle that allows flow, as depicted in FIGS. **11** and **12**, or it could just be anything that when wedged into the seal **254** prevents sealing. As such, the cannula **114** could include a thin solid rod to create a flow path around itself where the seal is poor, or a square profile might encourage even more flow. An open ended needle as depicted might be useful for puncturing a septum, for example, since a septum could be designed to reseal around the

perimeter of the needle. In some embodiments, a duck-billed valve configuration may seal well when un-interrupted against pressure in one direction.

(99) In some embodiments of the injector **100**, the reservoir **102** may be visible through the wall of the housing **122**. According to such embodiments, if the administration of the medical fluid or drug product is interrupted as a consequence of the operation of the lock **118**, the patient or caregiver may be able to visually determine the amount of the full dose remaining in the reservoir **102**. In other embodiments, the reservoir **102** may be completely enclosed within the housing **122** without provision for a window.

(100) To address the situation wherein it may be difficult or impossible to visualize the reservoir **102**, a third group of embodiments of the injector is illustrated in FIGS. **13-15**. This group of embodiments may be used in conjunction with any of the embodiments of the injector **100** described herein. These embodiments may be used with an injector with a reservoir **102** including a bore **104** having a first end **106** and a second end **108**, and a plunger assembly **110** including a plunger **112** moveable within the bore **104** between the first and second ends **106**, **108**. A cannula **114** may also be included, having an operational state wherein the cannula **114** is connected in fluid communication with the reservoir **102**. The injector **100** further includes a spring **116** coupled to the plunger assembly **110** to move the plunger **112** between the first and second ends **106**, **108**. All of these structures are illustrated in FIGS. **1-3**, for example, and are also applicable to the embodiment described relative to FIGS. **13-15**.

(101) In distinction to the previously described embodiments, the embodiment illustrated in FIGS. **13-15** includes an indicator **290** mechanically coupled to the plunger assembly **110**. The reservoir **102**, plunger arm **140**, and indicator **290** are at least partially disposed within a housing **122**.

(102) A proximity sensor **120** is coupled to the indicator **290** and moveable relative to the housing **122**, the proximity sensor **120** having a first sensor state wherein the proximity sensor **120** extends (e.g., extends fully) from the housing **122** and a second sensor state wherein the proximity sensor **120** is retracted toward the housing **122** relative to the first sensor state. In this version, the sensor **120** resembles the pivoting sensor **120** of FIGS. **4-6** and **9-10**, and as such, it should be appreciated that the indicator **290** could be implemented into the embodiments of FIGS. **4-6** and **9-10** or any other embodiments disclosed hereinabove. That is, the indicator **290** can be combined with other injectors with locks, as disclosed. In the version of FIGS. **14** and **15**, the indicator **290** is coupled to the plunger assembly **110** with the proximity sensor **120** in the second sensor state and the indicator **290** is decoupled from the plunger assembly **110** with the proximity sensor **120** in the first sensor state.

(103) The indicator **290** mirrors the internal motion of the plunger **112** externally to the housing **122** to permit visualization. Once the proximity sensor **120** detects that the injector **100** has become detached from the patient, the indicator **290** would be decoupled from the plunger **112**, such that the indicator **290** would reflect the portion of the dose remaining in the reservoir **102**. According to certain embodiments, the indicator **290** may be reversibly coupled to the plunger assembly **110** so that if the injector **100** permits the injection to resume after the injector **100** is repositioned or otherwise adjusted, the indicator **290** would continue to mirror the motion of the plunger assembly **110** when the injection is resumed. According to other embodiments, the indicator **290** may be irreversibly decoupled from the plunger assembly **110** so that the condition of the injector **100** when an error in administration occurred would be preserved. Preserving the condition of the injector **100** may be useful in assisting the patient or caregiver in their determination for future action based on the amount administered through the intended method of administration (i.e., prior to the occurrence) and the patient's overall disease/therapy conditions.

(104) For example, the plunger assembly **110** may include a plunger arm **140** attached to the plunger **112**, as illustrated in FIG. **14**. According to such an embodiment, the indicator **290** may be mechanically coupled to the plunger arm **140** through a gear train **292** having at least one gear **294** that is moveable into and out of engagement. The proximity sensor **120** may be coupled to the at

least one gear **294** to move the at least one gear **294** out of engagement and decouple the indicator **290** from the plunger arm **140** with the proximity sensor **120** in the first sensor state.

(105) According to the embodiment illustrated in FIGS. **14** and **15**, the gear train **292** includes the moveable pinion gear **294**, as well as two racks **296**, **298** formed on facing surfaces **300**, **302** of the plunger arm **140** and the indicator **290**. When the gear **294** engages the racks **296**, **298** simultaneously, as shown in FIG. **15**, the gear **294** transmits motion of the plunger arm **140** to the indicator **290**. When the gear **294** is disengaged from rack **296**, as shown in FIG. **14**, the indicator **290** is decoupled from any motion of the plunger arm **140**.

(106) The pinion gear **294** is biased toward engagement with the racks **296**, **298** by a spring **304**. A linkage **306** passes through the gear **294** at a first end **308**, and is attached to a further linkage **310** at a second end **312**. The linkage **310** may be a flexible linkage (as illustrated), in the form of a string or cord. In other embodiments, the linkage **310** may include a rigid mechanical linkage including additional levers, gears, couplings, etc. The depicted flexible linkage **310** may pass about a drum, wheel or pulley **314** and be connected to the proximity sensor **120**. The proximity sensor **120** may be defined by a lever **316** having a first end **318** attached to the housing **112** at a pivot **320** and a second **322** that is attached to the linkage **310**. The lever **316** may be biased to move clockwise about the pivot **320**, as illustrated in FIGS. **14** and **15**, by a further spring **324**.

(107) The injector **100** of any of the foregoing embodiments may be an on-body injector, the injector **100** comprising a surface having adhesive applied thereto to attach the injector **100** to a body of a patient. For example, FIGS. **16-18** illustrates an on-body injector **400** that may incorporate any of the embodiments illustrated in FIGS. **1-15**. In some embodiments, the injector **400** may be a wearable, disposable system. The injector **400** may include a disposable housing **402** that may be attached to a patient or wearer with adhesive, for example.

(108) The disposable housing **402** may be made of a plastic material. As seen in FIG. **17**, the housing **402** may be defined by two sections, a plate **404** that is applied against the wearer's skin, and a dome **406** that is attached to the plate **404**, preferably by a seal at an interface between a peripheral edge **408** of the plate **404** and a peripheral edge **410** of the dome **406**.

(109) As shown in FIG. **17**, the housing **402** has an interior surface **412** defining an interior space **414**, and an exterior surface **416**. In particular, the plate **404** has an interior surface **418** and an exterior surface **420**, and the dome **406** has an interior surface **422** and an exterior surface **424**. According to the illustrated embodiment, the interior surface **412** of the housing **402** is defined by the interior surfaces **418**, **422** of the plate **404** and the dome **406**, while the exterior surface **416** of the housing **402** is defined by the exterior surfaces **420**, **424** of the plate **404** and dome **406**.

(110) The housing **402** may be attached to the skin of the wearer. In some embodiments, an adhesive may be used. The adhesive may be adapted to releasably secure the housing to skin during a single application. As shown in FIG. **16**, the adhesive is disposed in a layer **426** on a portion **428** of the exterior surface **416** of the housing **402**, and in particular on the exterior surface **420** of the plate **404**. The adhesive is covered with a removable, disposable sheet **430** prior to application of the housing **402** to the skin of the wearer.

(111) As seen in FIGS. **17** and **18**, a reservoir **440**, a drive **442**, a cannula (or structure, see below) **444**, and an inserter **446** are disposed in the housing **402**. According to the illustrated embodiment, the reservoir **440** may be defined at least in part by a combination of a rigid-walled cylinder or bore **450** having a port **452** at a first end **454** and a plunger **456** fitted to move along a longitudinal axis **458** of the cylinder **450** between a second end **460** and the first end **454** to force medical fluid or drug product out of the reservoir **440** through the port **452** (FIG. **17**). The movement of the plunger **456** may be caused by the operation of the drive **442**. Additionally, as shown, a valve **451** can be disposed at the exit of the port **452** between the reservoir **440** and a delivery line **453** that is fluidly coupled to the cannula **444**.

(112) The drive **442** may include a plunger arm and a spring. The plunger arm may be in contact at least at a first end of the plunger **456** to urge the plunger **456** along the cylinder **450**, and the spring

may be coupled to the plunger arm to cause the plunger arm/plunger **456** to move along the longitudinal axis **458**. The combination of the plunger arm and spring may also be referred to as one example of a drive or an actuator. Other mechanisms, such as pressurized gases, materials undergoing phase changes and the like, may be used as the drive **442** to apply a force to the plunger **456** to move the plunger **456** along the cylinder **450**. To this extent, the drive **442** may be constructed according to the embodiments disclosed above relative to FIGS. **1-12**, or any other embodiment capable of serving the intended purpose.

(113) According to other embodiments, a non-rigid collapsible pouch may be substituted for the rigid-walled cylinder **450** and the plunger **456** illustrated in FIG. **17**. In embodiments where the reservoir **450** is in the form of a non-rigid collapsible pouch, a spring-based mechanical system may be used to compress and pressurize the reservoir **450**. In other embodiments, a non-mechanical system may be used to move the plunger **456** or compress the non-rigid collapsible reservoir. For example, a gas-generating system may be used, including a two-component system wherein the components are kept apart until the gas is to be generated, in which case they are combined. In some embodiments, a swellable gel may be used, wherein the introduction of water from a source internal to the device causes the gel to increase in dimension to move the plunger or compress the reservoir. As a further example, a propellant reservoir may be opened and the propellant discharged to move the plunger **456** or compress the reservoir. Embodiments of the present disclosure could be used to prevent motion of the structures used to move the plunger **456** or compress the reservoir, thereby limiting such motion if the injector **400** were to separate from the patient during delivery.

(114) According to certain embodiments, the reservoir **440** may be a pre-filled container, such as a pre-filled cartridge or a pre-filled syringe. Alternatively, the injector **400** may include a fill port **480** in fluid communication with the reservoir **440**, the fill port **480** adapted to receive a luer tip of a syringe, although a rubber septum may be used instead, for example. In use, a healthcare provider may inject the medical fluid or drug product from the syringe through the fill port **480** into the reservoir **440**, and the syringe may be provided as a pre-filled syringe (filled with any of the materials mentioned above) to the healthcare provider with the delivery injector **400** as a kit.

(115) The cannula (or structure) **444** may have a retracted state wherein a pointed end **490** (or, the entire cannula **444**) may be withdrawn inside the housing **402** and a deployed state wherein the pointed end **490** projects from the housing **402**, the inserter **446** moving the cannula (or structure) **444** from the retracted state to the deployed state. In some embodiments, the injector **400** includes a proximity sensor **120** similar to any one of the proximity sensors **1210** described above and which may extend from the housing **402** (specifically, the plate **404**) in a similar fashion to the cannula **444**, moving between extended and retracted states.

(116) The cannula **444** may be hollow, and may be used to administer the medical fluid or drug product directly to the patient. Alternatively, the structure **444** may be used in conjunction with a cannula **492** fitted about the structure **444**, the structure **444** being used to insert the cannula **492** into the patient through the injection site, and the medical fluid or drug product passing through the catheter **492** into the patient during administration. The injector **400** may, according to certain exemplary embodiments, use a needle to automatically insert a soft cannula into the subcutaneous tissue with the needle being withdrawn prior to passage of medical fluid or drug product through the soft cannula.

(117) As illustrated in FIG. **17**, the housing **402** (specifically the plate **404**) may have an aperture or opening **494** to permit the cannula (or structure) **444** (and optionally cannula **492**) to pass therethrough. According to certain embodiments, the aperture **494** may be unobstructed, such that there is no impediment or obstacle to the movement of the cannula **444** (and catheter **492**) through the opening **494**. However, to better maintain the sterility of the cannula **444** and the device's container closure integrity (CCI), a septum may be disposed in or over the aperture **494**.

(118) The septum, which may be made of a rubber, may be disposed between the cannula **444** (and the space **414**) and the patient's skin with the needle **444** in the retracted state. In the deployed state,

at least a portion of the needle **444** (i.e., the pointed end **490**) will extend from the space **414** through the septum. As such, the septum is always present as a barrier between the interior space **414** and the external environment.

(119) In some embodiments, the injector **400** includes a controller **500**. The controller may include at least one processor and memory, the processor programmed to carry out the actions that the controller is adapted to perform, and the memory, including one or more tangible non-transitory readable memories, having executable instructions stored thereon, which instructions when executed by the at least one processor may cause the at least one processor to carry out the actions that the controller is adapted to perform. In other embodiments, the controller may include circuitry that carries out the actions that the controller is adapted to perform. By way of example and not limitation, the controller **500** may be adapted to carry out any one of the methods described above relative to the drug delivery system.

(120) According to certain embodiments of the present disclosure, the lock and proximity sensor may be defined by a single structure, such as a plate or lever, or an assembly of structures, which collectively provide the function. Alternatively, the lock may be a structure defined separately from the proximity sensor. In such a case, the movement of the proximity sensor may be detected through the use of a switch, such as a mechanical switch or optical switch, which switch may be coupled to the controller **500** and may provide information in the form of a signal (or signals, or an absence of signal (low vs. high)) to the controller **500** when the proximity sensor moves or changes positions or states. In some embodiments, the lock can be the valve **451** connected to the controller **500**, as depicted in FIGS. **17** and **18**, and the proximity sensor can include an electrical sensor (e.g., a Hall effect sensor) or an electromechanical sensor (e.g., a linear variable differential transformer (LVDT)), for example, mounted within the housing for detecting the position of the movable cannula **444** or other movable proximity sensor mounted to the housing. The controller **500** may thus be operatively coupled to the valve **451**, and may be capable of transmitting information to the valve **451** in the form of a signal for changing the state of the valve **451** between positions or states in response to the signal received from the switch or other device coupled to the cannula **444** or other proximity sensor. For example, the controller **500** may be coupled to an actuator, such as a linear actuator, that is capable of moving the valve **451** between opened and closed positions or states.

(121) In some embodiments, other advantages of using the controller **500** can include defining the state of the lock or valve **451** as reversible or irreversible. For example, in some embodiments, the proximity sensor may have not only first and second states, but also one or more third states (e.g., intermediate states), where the lock (e.g., valve **451**) may transition from occupying a reversible condition to an irreversible condition. This functionality could be therefore achieved through the use of the controller **500** in operable communication with the proximity sensor and lock (e.g., valve **451**). In this manner, the sensitivity or threshold for converting the lock condition from reversible to irreversible could be specifically managed by way of the logic or software stored on a memory of the controller **500** based on the specific patient, the specific medical fluid or drug product being administered, or generally any other parameter.

(122) While the lock in FIGS. **17** and **18** has been described as a valve **451**, in other embodiments, the lock can take other forms. For example, FIGS. **21** and **22** depict an alternative version of the injector **400**, wherein the lock includes a movable plate **461** similar to the proximity sensor **120** described with respect to FIGS. **1-3**, for example. The plate **461** is spring biased into the position illustrated in FIG. **21** such that similar to prior embodiments, upon application of the injector **400** to a patient's skin, the plate **461** retracts into the housing **402** and into the position depicted in FIG. **22**. As shown, the plate **461** also includes an aperture **465** through which the fluid line **453** of the injector **400** passes. In FIGS. **21** and **22**, the fluid line **453** is a flexible line. And, in FIG. **21**, an interior edge **463** of the aperture **465** in the plate **461** bears down against an upper sidewall of the fluid line **453** such that the plate **461** pinches the fluid line closed. In this manner, the interior edge

463 of the aperture **465** in the plate **461** limits or prevents fluid in the reservoir **440** from travelling past the pinched portion of the fluid line **453**. The plate **461** thus serves as the lock. When the injector **400** is applied to a patient's skin, the plate **461** retracts sufficiently far into the housing that the interior edge **463** of the aperture **465** moves out of engagement with the fluid line, as shown in FIG. 22, such that medical fluid or drug product stored in the reservoir **440** can freely pass through the fluid line **453** under the operation of the drive **442**.

(123) The injector of any of the foregoing embodiments may be a hand-held autoinjector **600**, the injector comprising a needle shield that defines the proximity sensor. See FIGS. 19 and 20. The autoinjector **600** may include a housing **610** in which may be disposed assemblies or structures that insert or enable insertion of a cannula into the patient, and that inject a medical fluid or drug product from the reservoir through the cannula into the patient. According to certain embodiments, the same assemblies or structures that insert the cannula into the patient may also allow flow of the medical fluid or drug product from the reservoir through the cannula into the patient. The autoinjector **600** may also include assemblies or structures that connect the cannula to the reservoir, that withdraw the cannula into the housing **610**, or that deploy other structures that will prevent contact with the cannula once the cannula has been removed from the patient. The specific embodiment of the autoinjector **600** discussed below is provided by way of example and not of limitation. In some embodiments, the autoinjector **600** may or may not include assemblies or structures that direct or force insertion of the cannula (e.g., needle) into the patient. In some embodiments, the cannula may be substantially fixed relative to the housing of the autoinjector **600** such that insertion of the cannula into the patient simply results from moving the cannula into contact with the patient.

(124) In some embodiments, the drug delivery system **600** includes a reservoir **612** and a cannula **614** having a first end **616** that may be connected or connectable in fluid communication to the reservoir **612** and a second end **618** that may be inserted into a patient. The cannula **614** may be, for example, a rigid needle having a beveled edge that may be sized such that the second end **618** of the needle **614** is received under the skin so as to deliver a subcutaneous injection of the medical fluid or drug product within the reservoir **612**. The first end **616** of the needle **614** may be disposed through a wall **620** of the reservoir **612**, and thus be connected in fluid communication with the reservoir **612**. As illustrated, the first end **616** of the needle **614** may be disposed only partially through the wall **620** (which wall **620** may be a resealable septum or stopper, for example) such that the first end **616** of the needle **614** may not be connected in fluid communication with the reservoir **612** until the second end **618** of the needle **614** is inserted into the patient. So configured, the wall **620** of this embodiment is a lock, as used herein, and as will be described. Moreover, in some embodiments, once removed from the patient, the first end **616** of the needle **614** may again become disconnected from fluid communication with the reservoir **612**. In such a circumstance, the first end **616** of the needle **614** may be described as connectable in fluid communication with the reservoir **612**, although it will be recognized that there are other mechanisms by which the first end **616** of the needle **614** may be connectable, but not connected, in fluid communication with the reservoir **612**.

(125) In some embodiments, the drug delivery device **600** includes a shield **622** that may be deployed at least after the injection has been completed to limit access to the second end **618** of the needle **614**. According to certain embodiments, the shield **622** may have a biasing element (such as a spring) that extends the shield **622** from the housing **610** such that a distal end **626** of the shield **622** extends beyond the second end **618** of the needle **614** except when the shield **622** is disposed against the skin and the injection of the needle **614** is actuated. In fact, the injection of the needle **614** may be actuated according to certain embodiments of the autoinjector **600** by disposing the distal end **626** of the shield **622** on or against the skin of the patient. According to certain embodiments of the present disclosure, the shield **622** may define the proximity sensor **120**.

(126) In some embodiments, the drug delivery device **600** includes at least one drive **630** that may

be used to insert the second end **618** of the needle **614** into the skin of the patient, and to inject the medical fluid or drug product from the reservoir **612** through the needle **614** into the patient. The drive **630** may include one or more springs, according to certain embodiments. In other embodiments, the drive **630** may include a source of pressurized gas or a source of a material that undergoes a phase change, such that the escaping gas or phase changing material that provides a motive force that may be applied to the reservoir **612** to eject the medical fluid or drug product therefrom. According to still other embodiments, the drive **630** may include an electromechanical system, such as a motor.

(127) The drive **630** may cooperate with a wall **632** of the reservoir **622** to move that wall **632** toward the patient's skin. In accordance with such an embodiment, the wall **632** may be a stopper that is received within a bore **634**, and which may move along the bore **634** from a first end to a second end to inject the medical fluid or drug product from the reservoir **612**. The drive **630** may also cooperate with the wall **632** and/or the bore **634** to move the reservoir **612** relative to the housing **610** so as to move the second end **618** of the needle **614** relative to the housing **610** and into the patient. According to those embodiments wherein the drive **630** cooperates with the stopper **632**, this may occur before the first end **616** of the needle **614** is in fluid communication with the reservoir **612**. According to those embodiments wherein the drive cooperates with the bore **634**, the drive may include one component (e.g., first spring) that cooperates with the bore **634** to move the reservoir **612** and needle **614** relative to the housing **610**, and a second component (e.g., second spring) that cooperates with the stopper **632** to move the stopper **632** relative to the bore **634**.

(128) In some embodiments, the drive **630** is associated with an actuator **640**. The actuator **640** activates the drive to cause the drive **630** to insert the needle **614** and inject the medical fluid or drug product from the reservoir **612** through the needle **614** into the patient. The actuator **640** may, according to certain embodiments, be the shield **622**. According to other embodiments, the actuator **640** may be a button that may be depressed by the user once the autoinjector **600** is disposed on or against the patient's skin. While the embodiment illustrated in FIGS. **19** and **20** has the actuator **640** disposed at one end of the device, the actuator **640** could be disposed on the side of the device.

(129) According to certain embodiments, the autoinjector **600** may include at least one processor and memory. The controller may also include or be coupled to a power supply, e.g. a battery. The processor may be programmed to carry out the actions that the controller is adapted to perform and the memory may include one or more tangible non-transitory readable memories having executable instructions stored thereon, which instructions when executed by the at least one processor may cause the at least one processor to carry out the actions that the controller is adapted to perform. Alternatively, the controller may include other circuitry that carries out the actions that the controller is adapted to perform.

(130) Before the first end **616** penetrates the wall **620**, the wall **620** seals the reservoir **612** from communicating with the needle **614** and thereby limits the movement of the wall **632** or stopper in the bore **634**. In this way, the wall **620** is a lock, as used herein. But as mentioned above, as the second end **618** of the needle **614** of the autoinjector **600** of FIGS. **19** and **20** is inserted into a patient, the first end **616** of the needle **614** penetrates the wall **620** to become in fluid communication with the reservoir **612**. In some embodiments, once the second end **618** of the needle **614** is removed from the patient, the first end **616** of the needle **614** may again become disconnected from fluid communication with the reservoir **612** by, for example, being drawn back through the wall **620** at least such that the first end **616** of the needle **614** no longer resides within the reservoir **612**, but rather resides inside of the wall **620** or, said another way, resides at a point between opposing surfaces of the wall **620**. In this way, the wall **620** can automatically seal the opening through which the first end **616** of the needle **614** previously passed into the reservoir, and thereby locks the fluid delivery mechanism.

(131) FIGS. **23** and **24** illustrate an alternative version of the autoinjector **600** depicted in FIGS. **19** and **20** for facilitating this functionality. In FIGS. **23** and **24**, the autoinjector **600** includes a needle

assembly **700** for disconnecting the needle **614** from fluid communication with the reservoir **612** when the needle **614** is removed from the patient. As illustrated, the needle assembly **700** includes a retainer **702** and a biasing member such as spring **704**. The retainer **702** is fixed to the needle **614** and slidably disposed in the housing **610** adjacent to the shield **622**. The spring **704** can be a coil spring and is disposed between the wall **620** of the reservoir **612** and the retainer **702**. In the depicted version, a top surface **706** of the retainer **702** defines an annular recess **708** into which a first end **710** of the spring **704** is seated. A second end **712** of the spring **704** is seated against the wall **620** (or a component surrounding the wall **620**). So configured, the spring **704** biases the needle assembly **700** away from the reservoir **612** and the wall **620** (i.e., downward relative to the orientation of the autoinjector **600** depicted in FIG. 23).

(132) Accordingly, when the second end **618** of the needle **614** is inserted into a patient, a bottom surface **714** of the retainer **702** bears against the patient's skin and therefore serves as the proximity sensor in this embodiment. As a force is applied to the autoinjector **600** in the direction of the patient's skin, the needle assembly **700** moves into the housing **610** against the force of the spring **704** such that the first end **616** of the needle **614** fully penetrates the wall **620** and becomes in fluid communication with the reservoir **612**. This allows medicament in the reservoir **612** to pass into the first end **616** of the needle **614** for patient administration. When the needle **614** is removed from the patient, the spring **702** naturally biases the retainer **702** back into the position depicted in FIG. 23 which in turn draws the needle **614** with it because the needle **614** and retainer **702** are fixed together. In the version shown in FIG. 23, the housing **610** of the autoinjector **600** further includes one or more stops **716** carried by the inside of the housing **610** adjacent to the shield **622**. As the spring **702** biases the retainer **702** away from the reservoir **612**, the retainer **702** comes into contact with the stop(s) **716**. The stop(s) **716** therefore limit the displacement of the retainer **702** and needle **614** away from the reservoir **612** such that when the retainer **702** engages the stop(s) **716**, the first end **616** of the needle **614** is fully withdrawn from and out of fluid communication with the reservoir **612**. In the depicted version, this also means that the first end **616** of the needle **614** resides within the wall **620**, which serves as the lock. That is, the wall **620** in this configuration limits further movement of the wall **623** or stopper by sealing the flow path through the needle **614** when the device is removed from the patient. In this version, the wall **620**, as the lock, is reversible because when the second end **618** of the needle **614** is subsequently re-inserted into a patient, for example, the needle assembly **700** can again move into the housing **610** against the force of the spring **704** such that the first end **616** of the needle **614** fully penetrates the wall **620** and becomes in fluid communication with the reservoir **612**. In other versions, the device may include additional structures that prevent such reversible action from occurring.

(133) The present disclosure describes various systems and methods for use with a drug delivery device. The system, methods or drug delivery device can comprise use of a medical fluid or drug product listed below, with the caveat that the following list should neither be considered to be all inclusive nor limiting. The medical fluid or drug product will be contained in a reservoir. In some instances, the reservoir is a primary container that is either filled or pre-filled for treatment with the medical fluid or drug product. The primary container can be a cartridge or a pre-filled syringe.

(134) For example, the drug delivery device or more specifically the reservoir of the device may be filled with colony stimulating factors, such as granulocyte colony-stimulating factor (G-CSF). Such G-CSF agents include, but are not limited to, Neupogen® (filgrastim) and Neulasta® (pegfilgrastim). In various other embodiments, the drug delivery device may be used with various pharmaceutical products, such as an erythropoiesis stimulating agent (ESA), which may be in a liquid or a lyophilized form. An ESA is any molecule that stimulates erythropoiesis, such as Epogen® (epoetin alfa), Aranesp® (darbepoetin alfa), Dynepo® (epoetin delta), Mircera® (methoxy polyethylene glycol-epoetin beta), Hematide®, MRK-2578, INS-22, Retacrit® (epoetin zeta), Neorecormon® (epoetin beta), Silapo® (epoetin zeta), Binocrit® (epoetin alfa), epoetin alfa Hexal, Abseamed® (epoetin alfa), Ratioepo® (epoetin theta), Eporatio® (epoetin theta), Biopoin®

(epoetin theta), epoetin alfa, epoetin beta, epoetin zeta, epoetin theta, and epoetin delta, as well as the molecules or variants or analogs thereof as disclosed in the following patents or patent applications: U.S. Pat. Nos. 4,703,008; 5,441,868; 5,547,933; 5,618,698; 5,621,080; 5,756,349; 5,767,078; 5,773,569; 5,955,422; 5,986,047; 6,583,272; 7,084,245; and 7,271,689; and PCT Publ. Nos. WO 91/05867; WO 95/05465; WO 96/40772; WO 00/24893; WO 01/81405; and WO 2007/136752.

(135) An ESA can be an erythropoiesis stimulating protein. As used herein, “erythropoiesis stimulating protein” means any protein that directly or indirectly causes activation of the erythropoietin receptor, for example, by binding to and causing dimerization of the receptor. Erythropoiesis stimulating proteins include erythropoietin and variants, analogs, or derivatives thereof that bind to and activate erythropoietin receptor; antibodies that bind to erythropoietin receptor and activate the receptor; or peptides that bind to and activate erythropoietin receptor. Erythropoiesis stimulating proteins include, but are not limited to, epoetin alfa, epoetin beta, epoetin delta, epoetin omega, epoetin iota, epoetin zeta, and analogs thereof, pegylated erythropoietin, carbamylated erythropoietin, mimetic peptides (including EMP1/hematide), and mimetic antibodies. Exemplary erythropoiesis stimulating proteins include erythropoietin, darbepoetin, erythropoietin agonist variants, and peptides or antibodies that bind and activate erythropoietin receptor (and include compounds reported in U.S. Publ. Nos. 2003/0215444 and 2006/0040858) as well as erythropoietin molecules or variants or analogs thereof as disclosed in the following patents or patent applications: U.S. Pat. Nos. 4,703,008; 5,441,868; 5,547,933; 5,618,698; 5,621,080; 5,756,349; 5,767,078; 5,773,569; 5,955,422; 5,830,851; 5,856,298; 5,986,047; 6,030,086; 6,310,078; 6,391,633; 6,583,272; 6,586,398; 6,900,292; 6,750,369; 7,030,226; 7,084,245; and 7,217,689; US Publ. Nos. 2002/0155998; 2003/0077753; 2003/0082749; 2003/0143202; 2004/0009902; 2004/0071694; 2004/0091961; 2004/0143857; 2004/0157293; 2004/0175379; 2004/0175824; 2004/0229318; 2004/0248815; 2004/0266690; 2005/0019914; 2005/0026834; 2005/0096461; 2005/0107297; 2005/0107591; 2005/0124045; 2005/0124564; 2005/0137329; 2005/0142642; 2005/0143292; 2005/0153879; 2005/0158822; 2005/0158832; 2005/0170457; 2005/0181359; 2005/0181482; 2005/0192211; 2005/0202538; 2005/0227289; 2005/0244409; 2006/0088906; and 2006/0111279; and PCT Publ. Nos. WO 91/05867; WO 95/05465; WO 99/66054; WO 00/24893; WO 01/81405; WO 00/61637; WO 01/36489; WO 02/014356; WO 02/19963; WO 02/20034; WO 02/49673; WO 02/085940; WO 03/029291; WO 2003/055526; WO 2003/084477; WO 2003/094858; WO 2004/002417; WO 2004/002424; WO 2004/009627; WO 2004/024761; WO 2004/033651; WO 2004/035603; WO 2004/043382; WO 2004/101600; WO 2004/101606; WO 2004/101611; WO 2004/106373; WO 2004/018667; WO 2005/001025; WO 2005/001136; WO 2005/021579; WO 2005/025606; WO 2005/032460; WO 2005/051327; WO 2005/063808; WO 2005/063809; WO 2005/070451; WO 2005/081687; WO 2005/084711; WO 2005/103076; WO 2005/100403; WO 2005/092369; WO 2006/50959; WO 2006/02646; and WO 2006/29094.

(136) Examples of other pharmaceutical products for use with the device may include, but are not limited to, antibodies such as Vectibix® (panitumumab), Xgeva™ (denosumab) and Prolia™ (denosumab); other biological agents such as Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker), Neulasta® (pegfilgrastim, pegylated filgrastim, pegylated G-CSF, pegylated hu-Met-G-CSF), Neupogen® (filgrastim, G-CSF, hu-MetG-CSF), and Nplate® (romiplostim); small molecule drugs such as Sensipar® (cinacalcet). The device may also be used with a therapeutic antibody, a polypeptide, a protein or other chemical, such as an iron, for example, ferumoxytol, iron dextrans, ferric glyconate, and iron sucrose. The pharmaceutical product may be in liquid form, or reconstituted from lyophilized form.

(137) Among particular illustrative proteins are the specific proteins set forth below, including fusions, fragments, analogs, variants or derivatives thereof:

(138) OPGL specific antibodies, peptibodies, and related proteins, and the like (also referred to as

RANKL specific antibodies, peptibodies and the like), including fully humanized and human OPGL specific antibodies, particularly fully humanized monoclonal antibodies, including but not limited to the antibodies described in PCT Publ. No. WO 03/002713, as to OPGL specific antibodies and antibody related proteins, particularly those having the sequences set forth therein, particularly, but not limited to, those denoted therein: 9H7; 18B2; 2D8; 2E11; 16E1; and 22B3, including the OPGL specific antibodies having either the light chain of SEQ ID NO: 2 as set forth therein in FIG. 2 and/or the heavy chain of SEQ ID NO:4, as set forth therein in FIG. 4;

(139) Myostatin binding proteins, peptibodies, and related proteins, and the like, including myostatin specific peptibodies, particularly those described in US Publ. No. 2004/0181033 and PCT Publ. No. WO 2004/058988, including but not limited to peptibodies of the mTN8-19 family, including those of SEQ ID NOS: 305-351, including TN8-19-1 through TN8-19-40, TN8-19 con1 and TN8-19 con2; peptibodies of the mL2 family of SEQ ID NOS: 357-383; the mL15 family of SEQ ID NOS: 384-409; the mL17 family of SEQ ID NOS: 410-438; the mL20 family of SEQ ID NOS: 439-446; the mL21 family of SEQ ID NOS: 447-452; the mL24 family of SEQ ID NOS: 453-454; and those of SEQ ID NOS: 615-631;

(140) IL-4 receptor specific antibodies, peptibodies, and related proteins, and the like, particularly those that inhibit activities mediated by binding of IL-4 and/or IL-13 to the receptor, including those described in PCT Publ. No. WO 2005/047331 or PCT Appl. No. PCT/US2004/37242 and in US Publ. No. 2005/112694, particularly such antibodies as are described therein, particularly, and without limitation, those designated therein: L1H1; L1H2; L1H3; L1H4; L1H5; L1H6; L1H7; L1H8; L1H9; L1H10; L1H11; L2H1; L2H2; L2H3; L2H4; L2H5; L2H6; L2H7; L2H8; L2H9; L2H10; L2H11; L2H12; L2H13; L2H14; L3H1; L4H1; L5H1; L6H1;

(141) Interleukin 1-receptor 1 ("IL1-R1") specific antibodies, peptibodies, and related proteins, and the like, including but not limited to those described in U.S. Publ. No. 2004/097712A1, in parts pertinent to IL1-R1 specific binding proteins, monoclonal antibodies in particular, especially, without limitation, those designated therein: 15CA, 26F5, 27F2, 24E12, and 10H7, as disclosed in the aforementioned U.S. publication;

(142) Ang2 specific antibodies, peptibodies, and related proteins, and the like, including but not limited to those described in PCT Publ. No. WO 03/057134 and U.S. Publ No. 2003/0229023, particularly in parts pertinent to Ang2 specific antibodies and peptibodies and the like, especially those of sequences described therein and including but not limited to: L1(N); L1(N) WT; L1(N) 1K WT; 2xL1(N); 2xL1(N) WT; Con4 (N), Con4 (N) 1K WT, 2xCon4 (N) 1K; L1C; L1C 1K; 2xL1C; Con4C; Con4C 1K; 2xCon4C 1K; Con4-L1 (N); Con4-L1C; TN-12-9 (N); C17 (N); TN8-8(N); TN8-14 (N); Con 1 (N), also including anti-Ang 2 antibodies and formulations such as those described in PCT Publ. No. WO 2003/030833, particularly Ab526; Ab528; Ab531; Ab533; Ab535; Ab536; Ab537; Ab540; Ab543; Ab544; Ab545; Ab546; A551; Ab553; Ab555; Ab558; Ab559; Ab565; AbF1AbFD; AbFE; AbFJ; AbFK; AbG1D4; AbGC1E8; AbH1C12; Ab1A1; Ab1F; Ab1K, Ab1P; and Ab1P, in their various permutations as described therein, as disclosed in the foregoing publication;

(143) NGF specific antibodies, peptibodies, and related proteins, and the like including, in particular, but not limited to those described in US Publ. No. 2005/0074821 and U.S. Pat. No. 6,919,426, particularly as to NGF-specific antibodies and related proteins in this regard, including in particular, but not limited to, the NGF-specific antibodies therein designated 4D4, 4G6, 6H9, 7H2, 14D10 and 14D11, as disclosed in the foregoing publication;

(144) CD22 specific antibodies, peptibodies, and related proteins, and the like, such as those described in U.S. Pat. No. 5,789,554, as to CD22 specific antibodies and related proteins, particularly human CD22 specific antibodies, such as but not limited to humanized and fully human antibodies, including but not limited to humanized and fully human monoclonal antibodies, particularly including but not limited to human CD22 specific IgG antibodies, such as, for instance, a dimer of a human-mouse monoclonal hLL2 gamma-chain disulfide linked to a human-mouse

monoclonal hLL2 kappa-chain, including, but limited to, for example, the human CD22 specific fully humanized antibody in Epratuzumab, CAS registry number 501423-23-0;

(145) IGF-1 receptor specific antibodies, peptibodies, and related proteins, and the like, such as those described in PCT Publ. No. WO 06/069202, as to IGF-1 receptor specific antibodies and related proteins, including but not limited to the IGF-1 specific antibodies therein designated L1H1, L2H2, L3H3, L4H4, L5H5, L6H6, L7H7, L8H8, L9H9, L10H10, L11H11, L12H12, L13H13, L14H14, L15H15, L16H16, L17H17, L18H18, L19H19, L20H20, L21H21, L22H22, L23H23, L24H24, L25H25, L26H26, L27H27, L28H28, L29H29, L30H30, L31H31, L32H32, L33H33, L34H34, L35H35, L36H36, L37H37, L38H38, L39H39, L40H40, L41H41, L42H42, L43H43, L44H44, L45H45, L46H46, L47H47, L48H48, L49H49, L50H50, L51H51, L52H52, and IGF-1R-binding fragments and derivatives thereof;

(146) Also among non-limiting examples of anti-IGF-1R antibodies for use in the methods and compositions of the present disclosure are each and all of those described in: (i) US Publ. No. 2006/0040358 (published Feb. 23, 2006), 2005/0008642 (published Jan. 13, 2005), 2004/0228859 (published Nov. 18, 2004), including but not limited to, for instance, antibody 1A (DSMZ Deposit No. DSM ACC 2586), antibody 8 (DSMZ Deposit No. DSM ACC 2589), antibody 23 (DSMZ Deposit No. DSM ACC 2588) and antibody 18 as described therein; (ii) PCT Publ. No. WO 06/138729 (published Dec. 28, 2006) and WO 05/016970 (published Feb. 24, 2005), and Lu et al., 2004, J Biol. Chem. 279:2856-65, including but not limited to antibodies 2F8, A12, and IMC-A12 as described therein; (iii) PCT Publ. No. WO 07/012614 (published Feb. 1, 2007), WO 07/000328 (published Jan. 4, 2007), WO 06/013472 (published Feb. 9, 2006), WO 05/058967 (published Jun. 30, 2005), and WO 03/059951 (published Jul. 24, 2003); (iv) US Publ. No. 2005/0084906 (published Apr. 21, 2005), including but not limited to antibody 7C10, chimaeric antibody C7C10, antibody h7C10, antibody 7H2M, chimaeric antibody *7C10, antibody GM 607, humanized antibody 7C10 version 1, humanized antibody 7C10 version 2, humanized antibody 7C10 version 3, and antibody 7H2HM, as described therein; (v) US Publ. Nos. 2005/0249728 (published Nov. 10, 2005), 2005/0186203 (published Aug. 25, 2005), 2004/0265307 (published Dec. 30, 2004), and 2003/0235582 (published Dec. 25, 2003) and Maloney et al., 2003, Cancer Res. 63:5073-83, including but not limited to antibody EM164, resurfaced EM164, humanized EM164, huEM164 v1.0, huEM164 v1.1, huEM164 v1.32, and huEM164 v1.3 as described therein; (vi) U.S. Pat. No. 7,037,498 (issued May 2, 2006), US Publ. Nos. 2005/0244408 (published Nov. 30, 2005) and 2004/0086503 (published May 6, 2004), and Cohen, et al., 2005, Clinical Cancer Res. 11:2063-73, e.g., antibody CP-751,871, including but not limited to each of the antibodies produced by the hybridomas having the ATCC accession numbers PTA-2792, PTA-2788, PTA-2790, PTA-2791, PTA-2789, PTA-2793, and antibodies 2.12.1, 2.13.2, 2.14.3, 3.1.1, 4.9.2, and 4.17.3, as described therein; (vii) US Publ. Nos. 2005/0136063 (published Jun. 23, 2005) and 2004/0018191 (published Jan. 29, 2004), including but not limited to antibody 19D12 and an antibody comprising a heavy chain encoded by a polynucleotide in plasmid 15H12/19D12 HCA (74), deposited at the ATCC under number PTA-5214, and a light chain encoded by a polynucleotide in plasmid 15H12/19D12 LCF (κ), deposited at the ATCC under number PTA-5220, as described therein; and (viii) US Publ. No. 2004/0202655 (published Oct. 14, 2004), including but not limited to antibodies PINT-6A1, PINT-7A2, PINT-7A4, PINT-7A5, PINT-7A6, PINT-8A1, PINT-9A2, PINT-11A1, PINT-11A2, PINT-11A3, PINT-11A4, PINT-11A5, PINT-11A7, PINT-11A12, PINT-12A1, PINT-12A2, PINT-12A3, PINT-12A4, and PINT-12A5, as described therein, peptibodies, and related proteins and the like that target IGF-1 receptors;

(147) B-7 related protein 1 specific antibodies, peptibodies, related proteins and the like ("B7RP-1," also is referred to in the literature as B7H2, ICOSL, B7h, and CD275), particularly B7RP-specific fully human monoclonal IgG2 antibodies, particularly fully human IgG2 monoclonal antibody that binds an epitope in the first immunoglobulin-like domain of B7RP-1, especially those that inhibit the interaction of B7RP-1 with its natural receptor, ICOS, on activated T cells in

particular, especially, in all of the foregoing regards, those disclosed in U.S. Publ. No. 2008/0166352 and PCT Publ. No. WO 07/011941, including but not limited to antibodies designated therein as follow: 16H (having light chain variable and heavy chain variable sequences SEQ ID NO:1 and SEQ ID NO:7 respectively therein); 5D (having light chain variable and heavy chain variable sequences SEQ ID NO:2 and SEQ ID NO:9 respectively therein); 2H (having light chain variable and heavy chain variable sequences SEQ ID NO:3 and SEQ ID NO:10 respectively therein); 43H (having light chain variable and heavy chain variable sequences SEQ ID NO:6 and SEQ ID NO:14 respectively therein); 41H (having light chain variable and heavy chain variable sequences SEQ ID NO:5 and SEQ ID NO:13 respectively therein); and 15H (having light chain variable and heavy chain variable sequences SEQ ID NO:4 and SEQ ID NO:12 respectively therein);

(148) IL-15 specific antibodies, peptibodies, and related proteins, and the like, such as, in particular, humanized monoclonal antibodies, particularly antibodies such as those disclosed in U.S. Publ. Nos. 2003/0138421; 2003/023586; and 2004/0071702; and U.S. Pat. No. 7,153,507, as to IL-15 specific antibodies and related proteins, including peptibodies, including particularly, for instance, but not limited to, HuMax IL-15 antibodies and related proteins, such as, for instance, 146B7;

(149) IFN gamma specific antibodies, peptibodies, and related proteins and the like, especially human IFN gamma specific antibodies, particularly fully human anti-IFN gamma antibodies, such as, for instance, those described in US Publ. No. 2005/0004353, as to IFN gamma specific antibodies, particularly, for example, the antibodies therein designated 1118; 1118*; 1119; 1121; and 1121*. The entire sequences of the heavy and light chains of each of these antibodies, as well as the sequences of their heavy and light chain variable regions and complementarity determining regions, as disclosed in the foregoing US Publication and in Thakur et al., Mol. Immunol. 36:1107-1115 (1999). Specific antibodies include those having the heavy chain of SEQ ID NO: 17 and the light chain of SEQ ID NO:18; those having the heavy chain variable region of SEQ ID NO:6 and the light chain variable region of SEQ ID NO:8; those having the heavy chain of SEQ ID NO:19 and the light chain of SEQ ID NO:20; those having the heavy chain variable region of SEQ ID NO:10 and the light chain variable region of SEQ ID NO:12; those having the heavy chain of SEQ ID NO:32 and the light chain of SEQ ID NO:20; those having the heavy chain variable region of SEQ ID NO:30 and the light chain variable region of SEQ ID NO:12; those having the heavy chain sequence of SEQ ID NO:21 and the light chain sequence of SEQ ID NO:22; those having the heavy chain variable region of SEQ ID NO:14 and the light chain variable region of SEQ ID NO:16; those having the heavy chain of SEQ ID NO:21 and the light chain of SEQ ID NO:33; and those having the heavy chain variable region of SEQ ID NO:14 and the light chain variable region of SEQ ID NO:31, as disclosed in the foregoing US Publication. A specific antibody contemplated is antibody 1119 as disclosed in foregoing US Publication and having a complete heavy chain of SEQ ID NO:17 as disclosed therein and having a complete light chain of SEQ ID NO:18 as disclosed therein;

(150) TALL-1 specific antibodies, peptibodies, and the related proteins, and the like, and other TALL specific binding proteins, such as those described in U.S. Publ. Nos. 2003/0195156 and 2006/0135431, as to TALL-1 binding proteins, particularly the molecules of Tables 4 and 5B, as disclosed in the foregoing US Publications;

(151) Parathyroid hormone ("PTH") specific antibodies, peptibodies, and related proteins, and the like, such as those described in U.S. Pat. No. 6,756,480, particularly in parts pertinent to proteins that bind PTH;

(152) Thrombopoietin receptor ("TPO-R") specific antibodies, peptibodies, and related proteins, and the like, such as those described in U.S. Pat. No. 6,835,809, particularly in parts pertinent to proteins that bind TPO-R;

(153) Hepatocyte growth factor ("HGF") specific antibodies, peptibodies, and related proteins, and

the like, including those that target the HGF/SF:cMet axis (HGF/SF:c-Met), such as the fully human monoclonal antibodies that neutralize hepatocyte growth factor/scatter (HGF/SF) described in US Publ. No. 2005/0118643 and PCT Publ. No. WO 2005/017107, huL2G7 described in U.S. Pat. No. 7,220,410 and OA-5d5 described in U.S. Pat. Nos. 5,686,292 and 6,468,529 and in PCT Publ. No. WO 96/38557, particularly in parts pertinent to proteins that bind HGF;

(154) TRAIL-R2 specific antibodies, peptibodies, related proteins and the like, such as those described in U.S. Pat. No. 7,521,048, particularly in parts pertinent to proteins that bind TRAIL-R2;

(155) Activin A specific antibodies, peptibodies, related proteins, and the like, including but not limited to those described in US Publ. No. 2009/0234106, particularly in parts pertinent to proteins that bind Activin A;

(156) TGF-beta specific antibodies, peptibodies, related proteins, and the like, including but not limited to those described in U.S. Pat. No. 6,803,453 and US Publ. No. 2007/0110747, particularly in parts pertinent to proteins that bind TGF-beta;

(157) Amyloid-beta protein specific antibodies, peptibodies, related proteins, and the like, including but not limited to those described in PCT Publ. No. WO 2006/081171, particularly in parts pertinent to proteins that bind amyloid-beta proteins. One antibody contemplated is an antibody having a heavy chain variable region comprising SEQ ID NO: 8 and a light chain variable region having SEQ ID NO: 6 as disclosed in the International Publication;

(158) c-Kit specific antibodies, peptibodies, related proteins, and the like, including but not limited to those described in Publ. No. 2007/0253951, particularly in parts pertinent to proteins that bind c-Kit and/or other stem cell factor receptors;

(159) OX40L specific antibodies, peptibodies, related proteins, and the like, including but not limited to those described in U.S. application Ser. No. 11/086,289, particularly in parts pertinent to proteins that bind OX40L and/or other ligands of the OX40 receptor; and

(160) Other exemplary proteins, including Activase® (alteplase, tPA); Aranesp® (darbepoetin alfa); Epogen® (epoetin alfa, or erythropoietin); GLP-1, Avonex® (interferon beta-1a); Bexxar® (tositumomab, anti-CD22 monoclonal antibody); Betaseron® (interferon-beta); Campath® (alemtuzumab, anti-CD52 monoclonal antibody); Dynepo® (epoetin delta); Velcade® (bortezomib); MLN0002 (anti- $\alpha 4\beta 7$ mAb); MLN1202 (anti-CCR2 chemokine receptor mAb); Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker); Eprex® (epoetin alfa); Erbitux® (cetuximab, anti-EGFR/HER1/c-ErbB-1); Genotropin® (somatropin, Human Growth Hormone); Herceptin® (trastuzumab, anti-HER2/neu (erbB2) receptor mAb); Humatrope® (somatropin, Human Growth Hormone); Humira® (adalimumab); insulin in solution; Infergen® (interferon alfacon-1); Natrecor® (nesiritide; recombinant human B-type natriuretic peptide (hBNP)); Kineret® (anakinra); Leukine® (sargamostim, rhuGM-CSF); LymphoCide® (epratuzumab, anti-CD22 mAb); Benlysta™ (lymphostatin B, belimumab, anti-BlyS mAb); Metalyse® (tenecteplase, t-PA analog); Mircera® (methoxy polyethylene glycol-epoetin beta); Mylotarg® (gemtuzumab ozogamicin); Raptiva® (efalizumab); Cimzia® (certolizumab pegol, CDP 870); Soliris™ (eculizumab); pexelizumab (anti-C5 complement); Numax® (MEDI-524); Lucentis® (ranibizumab); Panorex® (17-1A, edrecolomab); Trabio® (lerdelimumab); TheraCim hR3 (nimotuzumab); Omnitarg (pertuzumab, 2C4); Osidem® (IDM-1); OvaRex® (B43.13); Nuvion® (visilizumab); cantuzumab mertansine (huC242-DM1); NeoRecormon® (epoetin beta); Neumega® (oprelvekin, human interleukin-11); Neulasta® (pegylated filgrastim, pegylated G-CSF, pegylated hu-Met-G-CSF); Neupogen® (filgrastim, G-CSF, hu-MetG-CSF); Orthoclone OKT3® (muromonab-CD3, anti-CD3 monoclonal antibody); Procrit® (epoetin alfa); Remicade® (infliximab, anti-TNF α monoclonal antibody); Reopro® (abciximab, anti-GP IIb/IIIa receptor monoclonal antibody); Actemra® (anti-IL6 Receptor mAb); Avastin® (bevacizumab), HuMax-CD4 (zanolimumab); Rituxan® (rituximab, anti-CD20 mAb); Tarceva® (erlotinib); Roferon-A® (interferon alfa-2a); Simulect® (basiliximab); Prexige® (lumiracoxib); Synagis® (palivizumab);

146B7-CHO (anti-IL15 antibody, see U.S. Pat. No. 7,153,507); Tysabri® (natalizumab, anti- α 4 integrin mAb); Valortim® (MDX-1303, anti-*B. anthracis* protective antigen mAb); ABthrax™; Vectibix® (panitumumab); Xolair® (omalizumab); ETI211 (anti-MRSA mAb); IL-1 trap (the Fc portion of human IgG1 and the extracellular domains of both IL-1 receptor components (the Type I receptor and receptor accessory protein)); VEGF trap (Ig domains of VEGFR1 fused to IgG1 Fc); Zenapax® (daclizumab); Zenapax® (daclizumab, anti-IL-2R α mAb); Zevalin® (ibritumomab tiuxetan); Zetia® (ezetimibe); Orencia® (atacept, TACI-Ig); anti-CD80 monoclonal antibody (galiximab); anti-CD23 mAb (lumiliximab); BR2-Fc (huBR3/huFc fusion protein, soluble BAFF antagonist); CNTO 148 (golimumab, anti-TNF α mAb); HGS-ETR1 (mapatumumab; human anti-TRAIL Receptor-1 mAb); HuMax-CD20 (ocrelizumab, anti-CD20 human mAb); HuMax-EGFR (zalutumumab); M200 (volociximab, anti-051 integrin mAb); MDX-010 (ipilimumab, anti-CTLA-4 mAb and VEGFR-1 (IMC-18F1); anti-BR3 mAb; anti-*C. difficile* Toxin A and Toxin B C mAbs MDX-066 (CDA-1) and MDX-1388); anti-CD22 dsFv-PE38 conjugates (CAT-3888 and CAT-8015); anti-CD25 mAb (HuMax-TAC); anti-CD3 mAb (NI-0401); adecatumumab; anti-CD30 mAb (MDX-060); MDX-1333 (anti-IFNAR); anti-CD38 mAb (HuMax CD38); anti-CD40L mAb; anti-Cripto mAb; anti-CTGF Idiopathic Pulmonary Fibrosis Phase I Fibrogen (FG-3019); anti-CTLA4 mAb; anti-eotaxin1 mAb (CAT-213); anti-FGF8 mAb; anti-ganglioside GD2 mAb; anti-ganglioside GM2 mAb; anti-GDF-8 human mAb (MYO-029); anti-GM-CSF Receptor mAb (CAM-3001); anti-HepC mAb (HuMax HepC); anti-IFN α mAb (MEDI-545, MDX-1103); anti-IGF1R mAb; anti-IGF-1R mAb (HuMax-Inflam); anti-IL12 mAb (ABT-874); anti-IL12/IL23 mAb (CNTO 1275); anti-IL13 mAb (CAT-354); anti-IL2Ra mAb (HuMax-TAC); anti-IL5 Receptor mAb; anti-integrin receptors mAb (MDX-018, CNTO 95); anti-IP10 Ulcerative Colitis mAb (MDX-1100); anti-LLY antibody; BMS-66513; anti-Mannose Receptor/hCG β mAb (MDX-1307); anti-mesothelin dsFv-PE38 conjugate (CAT-5001); anti-PD1mAb (MDX-1106 (ONO-4538)); anti-PDGFR α antibody (IMC-3G3); anti-TGF β mAb (GC-1008); anti-TRAIL Receptor-2 human mAb (HGS-ETR2); anti-TWEAK mAb; anti-VEGFR/Flt-1 mAb; anti-ZP3 mAb (HuMax-ZP3); NVS Antibody #1; and NVS Antibody #2.

(161) Also included can be a sclerostin antibody, such as but not limited to romosozumab, blosozumab, or BPS 804 (Novartis). Further included can be therapeutics such as rilotumumab, bixalomer, trebananib, ganitumab, conatumumab, motesanib diphosphate, brodalumab, vidupirant, panitumumab, denosumab, NPLATE, PROLIA, VECTIBIX or XGEVA. Additionally, included in the device can be a monoclonal antibody (IgG) that binds human Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9), e.g. U.S. Pat. No. 8,030,547, U.S. Ser. No. 13/469,032, WO2008/057457, WO2008/057458, WO2008/057459, WO2008/063382, WO2008/133647, WO2009/100297, WO2009/100318, WO2011/037791, WO2011/053759, WO2011/053783, WO2008/125623, WO2011/072263, WO2009/055783, WO2012/0544438, WO2010/029513, WO2011/111007, WO2010/077854, WO2012/088313, WO2012/101251, WO2012/101252, WO2012/101253, WO2012/109530, and WO2001/031007.

(162) Also included can be talimogene laherparepvec or another oncolytic HSV for the treatment of melanoma or other cancers. Examples of oncolytic HSV include, but are not limited to talimogene laherparepvec (U.S. Pat. Nos. 7,223,593 and 7,537,924); OncoVEXGALV/CD (U.S. Pat. No. 7,981,669); OrienX00 (Lei et al., 2013, World Journal of Gastroenterology, 19:5138-5143); G207, 1716; NV1020; NV12023; NV1034 and NV1042 (Vargehes et al. 2002, Cancer Gene Ther, 2002, 9 (12): 967-978).

(163) Also included are TIMPs. TIMPs are endogenous tissue inhibitors of metalloproteinases (TIMPs) and are important in many natural process. TIMP-3 is expressed by various cells or and is present in the extracellular matrix; it inhibits all the major cartilage-degrading metalloproteases, and may play a role in many degradative diseases of connective tissue, including rheumatoid arthritis and osteoarthritis, as well as in cancer and cardiovascular conditions. The amino acid sequence of TIMP-3, and the nucleic acid sequence of a DNA that encodes TIMP-3, are disclosed

in U.S. Pat. No. 6,562,596, issued May 13, 2003. Description of TIMP mutations can be found in U.S. 61/782,613, U.S. 61/798,160, U.S. 61/802,988, and US 61/940,67.

(164) Also included are antagonistic antibodies for human calcitonin gene-related peptide (CGRP) receptor and bispecific antibody molecule that target the CGRP receptor and other headache targets. Further information concerning these molecule can be found in WO2A075238A1.

(165) Additionally, a bispecific T cell engager antibody (BiTe), e.g. Blinotumomab can be used in the device. Alternatively, included can be an APJ large molecule agonist e.g., apelin or analogues thereof in the device. Information relating to such molecules can be found in PCT/2013/075773.

(166) Although the preceding text sets forth a detailed description of different embodiments of the disclosure, it should be understood that the legal scope of the invention is defined by the words of the claims set forth at the end of this patent. The detailed description is to be construed as exemplary only and does not describe every possible embodiment of the invention because describing every possible embodiment would be impractical, if not impossible. Numerous alternative embodiments could be implemented, using either current technology or technology developed after the filing date of this patent, that would still fall within the scope of the claims defining the invention.

(167) It should also be understood that, unless a term is expressly defined in this patent using the sentence “As used herein, the term ‘_____’ is hereby defined to mean . . .” or a similar sentence, there is no intent to limit the meaning of that term, either expressly or by implication, beyond its plain or ordinary meaning, and such term should not be interpreted to be limited in scope based on any statement made in any section of this patent (other than the language of the claims). To the extent that any term recited in the claims at the end of this patent is referred to in this patent in a manner consistent with a single meaning, that is done for sake of clarity only so as to not confuse the reader, and it is not intended that such claim term be limited, by implication or otherwise, to that single meaning. Finally, unless a claim element is defined by reciting the word “means” and a function without the recital of any structure, it is not intended that the scope of any claim element be interpreted based on the application of 35 U.S.C. § 112, sixth paragraph.

Claims

1. An on-body injector comprising: a reservoir filled or configured to be filled with a drug; a cannula; a drive operable to move the drug out of the reservoir; a proximity sensor having a first sensor state when the proximity sensor is not in contact with skin of a patient and a second sensor state when the proximity sensor is in contact with the skin of the patient, wherein the proximity sensor is configured to output at least one electrical signal as a result of changing from the first sensor state to the second sensor state; a lock having a first position to prevent the drive from moving the drug out of the reservoir when the proximity sensor is in the first sensor state and the lock having a second position permitting the drive to move the drug out of the reservoir when the proximity sensor is in the second sensor state, wherein the lock is operably coupled to the proximity sensor such that the proximity sensor changing from the first sensor state to the second sensor state causes the lock to move from the first position to the second position; and an actuator configured to move the lock between the first position and the second position.
2. The on-body injector of claim 1, comprising a housing, wherein the reservoir, cannula, and drive each is disposed at least partially within the housing.
3. The on-body injector of claim 2, comprising a controller disposed at least partially within the housing and being in communication with the proximity sensor and the lock.
4. The on-body injector of claim 3, the controller being configured to change the lock from the first position to the second position in response to the at least one electrical signal received from the proximity sensor in the second sensor state.
5. The on-body injector of claim 3, wherein the proximity sensor comprises an electrical sensor or

an electromechanical sensor.

6. The on-body injector of claim 3, wherein the controller comprises a processor and a memory.

7. The on-body injector of claim 2, wherein the cannula is moveable through an opening in the housing for insertion into the skin of the patient.

8. The on-body injector of claim 7, comprising an inserter disposed at least partially within the housing and configured to move the cannula through the opening for insertion into the skin of the patient.

9. The on-body injector of claim 2, comprising an indicator disposed at an exterior of the housing and configured to visually indicate progress of an injection.

10. The on-body injector of claim 1, wherein the cannula has a pre- operational state wherein the cannula is not connected in fluid communication with the reservoir.

11. The on-body injector of claim 1, comprising a plunger movably disposed in the reservoir, wherein the lock mechanically resists movement of the plunger between first and second ends of the reservoir when the proximity sensor is in the first sensor state.

12. The on-body injector of claim 1, wherein the drive comprises a motor.

13. The on-body injector of claim 1, wherein the drive comprises a source of pressurized gas.

14. The on-body injector of claim 1, comprising a controller configured to receive the at least one electrical signal output by the proximity sensor and control the lock.

15. An on-body injector comprising: a reservoir filled or configured to be filled with a drug; a cannula; a drive operable to move the drug out of the reservoir; a proximity sensor having a first sensor state when the proximity sensor is not in contact with skin of a patient and a second sensor state when the proximity sensor is in contact with the skin of the patient; a lock operably coupled to the proximity sensor, the lock having a first position to prevent the drive from moving the drug out of the reservoir when the proximity sensor is in the first sensor state and the lock having a second position permitting the drive to move the drug out of the reservoir when the proximity sensor is in the second sensor state; an actuator configured to move the lock between the first position and the second position; and a controller in communication with the proximity sensor, the lock, and the actuator, the controller being configured to change the lock from the first position to the second position in response to a signal received from the proximity sensor in the second sensor state.

16. The on-body injector of claim 15, comprising a housing having an opening, wherein the cannula is moveable through the opening in the housing for insertion into the skin of the patient.

17. The on-body injector of claim 16, comprising an inserter disposed at least partially within the housing and configured to move the cannula through the opening for insertion into the skin of the patient.

18. The on-body injector of claim 15, wherein the cannula has a pre-operational state wherein the cannula is not connected in fluid communication with the reservoir.
